THE SOLVABILITY DICHOTOMY FOR COMMUTATOR-TREE MIXING IN FINITE GROUPS

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Abstract. We study random products prescribed by balanced commutator trees, with independently configurable leaf labels and independent choices at every expanded occurrence. A tree is valid if its deterministic commutator value is non-identity. For a finite non-solvable group and a conjugation-closed generating leaf set, a uniform strict-contraction level exists exactly when every proper leaf-generated subgroup is solvable. Its exact value is the largest derived length of these subgroups. Among finite non-abelian simple groups, this property for every non-trivial conjugacy class characterises the minimal simple groups. We classify projective rank-one involution classes and their obstructions. An arbitrary-order stability estimate and bounded-rank diameter bounds give fixed-error mixing in optimal depth $\Theta(\log\log|G|)$ for all valid trees in the unobstructed rank-one families and all minimal simple groups. For primes $q \equiv \pm 3 \pmod 8$, $q \neq 3$, every pair of exterior S_4 subgroups of $\mathrm{PGL}_2(q)$ with intersection of order six is simultaneously conjugate to a natural Hamilton pair. Compression of a quaternionic Ramanujan walk bounds the largest fixed-space cosine over non-trivial irreducibles by $\sqrt{3}/2$, with asymptotic equality.

1. Introduction

Random products in a finite group provide a basic way to approach the uniform distribution. Each step may simply multiply by a prescribed group element or do nothing, with equal probability. We study what happens when the order of these steps is prescribed by repeatedly forming commutators $[x, y] = xyx^{-1}y^{-1}$. We require the underlying deterministic word to have non-identity value. Under this constraint, when does every allowed configuration begin to contract towards the uniform distribution, and what determines the required depth?

The answer is controlled by solvability. Iterated commutators in a solvable subgroup eventually become trivial, at a depth measured by its derived length. The non-identity requirement therefore prevents sufficiently deep configurations from remaining inside such a subgroup. We prove that, for a conjugation-closed set of allowed elements generating a finite non-solvable group, the exact contraction threshold is the largest derived length of a proper subgroup generated by allowed elements. If one of those proper subgroups is non-solvable, it instead supplies an obstruction at every depth.

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Equivalently, we ask when a non-identity iterated commutator certifies that its entries generate the whole group. Once the depth reaches the stated threshold, no proper subgroup generated by allowed entries can support such a value. This gives a group-theoretic interpretation of the contraction law.

Transpositions illustrate the distinction. In S_5, every proper subgroup generated by transpositions is solvable, and the largest derived length is 3. In S_n for $n \geq 6$, the point stabilizer S_{n-1} is already non-solvable: at every depth it supplies valid configurations supported in a proper subgroup. Thus uniform contraction holds for S_5 and is obstructed for every $n \geq 6$ (Corollary 5.1).

The model and the structural results. The leaves of a balanced binary tree are labelled by elements of a set $C \subseteq G$, and each internal vertex takes the commutator of its two child values. A configuration is *valid* when the root value is non-identity. In the random product obtained by expanding the tree, each occurrence is read independently: it contributes its prescribed element or the identity with equal probability. Thus repeated occurrences have fresh random choices; this is the *independent-read* model. The labels at different leaf positions may also be chosen independently, which we call the *configurable* model. Precise definitions are in Section 2.

A subgroup is C-generated if it is generated by its intersection with C. Strict contraction means that the Fourier operator has norm less than one in every non-trivial irreducible representation; a uniform level requires this for every valid configuration at that level. The first main theorem identifies this analytic threshold with a subgroup invariant.

Theorem A (Solvability dichotomy and exact level)**.** *Let G be a finite non-solvable group and C a configurable conjugation-closed leaf set with $\langle C \rangle = G$. The construction has a uniform strict-contraction level if and only if every proper C-generated subgroup of G is solvable; if some proper C-generated subgroup is non-solvable, the obstruction is permanent: every depth admits a valid configuration that fails to contract in some non-trivial irreducible. When the level exists, it is exactly $D = \max\{\mathrm{dl}(H) : H < G,\ H \text{ is } C\text{-generated}\}$, the largest derived length among proper C-generated subgroups* (Theorem 3.9).

The analytic ingredient concerns one configuration at a time. For a tree τ, let μ_τ be its unconditional output law and write $M_\tau(\rho) = \widehat{\mu_\tau}(\rho)$ for its Fourier operator. The subspace fixed by H is denoted by V_ρ^H. Throughout, norms of operators are operator norms, written $\|\cdot\|$.

Theorem B (Fourier fixed-vector criterion)**.** *For any unitary representation ρ and any configuration τ with leaf set S and $H = \langle S \rangle$, one has $\|M_\tau(\rho)\| = 1 \iff V_\rho^H \neq 0$. Hence the configuration contracts strictly in every non-trivial irreducible if and only if $\langle S \rangle = G$: strict contraction $\iff$ generation $\iff$ escape from every proper subgroup* (Lemma 3.1 and Theorem 3.2).

The common-fixed-vector argument proves Theorem B. The normal-subset word-value lemma of Fernández-Alcober and Pintonello [12, Lemma 2.1(ii)] identifies the subgroup generated by depth-k targets with $H^{(k)}$. Together they identify the

exact depth at which a non-identity target forces generation, proving Theorem A. Section 3.9 summarises this correspondence.

For involution leaves the averaging factors are orthogonal projections onto fixed spaces, connecting the criterion with the theory of products of projections. For leaves of arbitrary order, equality in the contraction inequality still characterises the fixed space. Angles between two subgroup-fixed spaces will govern the arithmetic application below.

Testing all non-trivial irreducibles matters: a proper subgroup can act transitively on a chosen permutation set, so contraction on its augmentation representation alone need not detect the obstruction. Proposition 3.4 explains this distinction; S_5 acting on six points supplies explicit examples.

For *rigid* templates—one fixed leaf per position, the regime of an actual branching program—the configurable level D becomes an upper bound, and persistence of validity is a finite dynamical question on conjugacy classes, which we solve.

Theorem C (Rigid templates)**.** *Let C be conjugation-closed.* (i) *For a non-empty set S of non-trivial conjugacy classes, valid binary compositions persist at every level from S if and only if some class in S has non-solvable normal closure—equivalently, escapes the solvable radical $O_\infty(G)$—if and only if some class in S reaches a cycle of the target transition graph $\mathcal{T}(G)$; and mixed-class children never extend the reach of persistence beyond conjugated doubling* (Theorem 5.12 and Corollaries 5.13 and 5.14). (ii) *Over the transposition class of S_5, the non-vacuously realisable exact levels of rigid families are precisely* $\{2, 3\}$ (Theorem 5.15).

Classification and examples. The subgroup criterion yields exact levels in concrete families. For involution leaves in A_5 the level is 2: although A_4 is a proper subgroup of derived length 2, its involutions generate only V_4, so only the subgroups generated by the allowed leaves enter the criterion (Corollary 5.2). The same criterion also characterises a classical family: a finite non-abelian simple group is minimal simple if and only if every non-trivial conjugacy class has a uniform strict-contraction level (Corollary 5.3). In that case every class has level at most 5, and the involution class of $\mathrm{PSL}_3(3)$ attains 5 (Proposition 6.4). The converse uses the solvable radical of a proper non-solvable subgroup; the bound uses the known bound on proper-subgroup derived lengths [11, Remark 2.3]. The transvection class in the same group has level 4: Proposition 4.2 classifies transvection leaves in $\mathrm{SL}_n(q)$ for $n \geq 3$, making the dependence on the leaf set explicit. For a generating conjugacy class in $\mathrm{GL}_2(\mathbb{Z}/p^r)$, $p \geq 5$, Proposition 4.4 transfers the dichotomy from reduction modulo p; at $r = 2$ the level increases by at most one.

In projective rank one the criterion gives a full involution-class classification. Here an *exterior* involution of $\mathrm{PGL}_2(q)$ is one whose determinant is a nonsquare; the other involutions lie in $\mathrm{PSL}_2(q)$.

Theorem D (Rank-one involutions)**.** *For odd $q \geq 5$, all involutions of $\mathrm{PGL}_2(q)$ are obstructed, whereas the exterior class is obstructed exactly by proper subfields of size at least 5 and odd extension degree. In the unobstructed exterior cases the exact level is 3 for $q \equiv \pm 3 \pmod 8$ and 2 for $q \equiv \pm 1 \pmod 8$. The involution classes of $\mathrm{PSL}_2(q)$, including its small exceptions, are classified in Theorem 6.2.*

Quantitative consequences and an arithmetic application. For a fixed group, strict contraction gives a positive gap by finiteness. Over a growing family, that gap may tend to zero. For involution leaves and uniformly bounded depth, a uniform contraction gap is equivalent to a uniform spectral gap for the associated lazy Cayley walks: each walk chooses an expanded leaf occurrence uniformly and applies it with probability $1/2$. Proposition 3.19 and Corollary 3.20 give explicit two-sided comparisons. This connects the block problem to expansion of its particular generating sets.

The next result controls the mixing depth even when the base gap decreases with group size. Write U_G for the uniform law on G.

Theorem E (Uniform rates and optimal order of depth)**.** *For the unobstructed classes of Theorem D and Theorem 6.2, and for every non-trivial conjugacy class of every minimal simple group, there are absolute $c, B > 0$ such that every valid tree τ of level $m \geq D$ satisfies*

$$d_{\mathrm{TV}}(\mu_\tau, U_G) \leq \tfrac{1}{2}\sqrt{|G|-1}\,\exp\bigl(-c4^m/(\log|G|)^B\bigr).$$

For fixed error in $(0, 1/2)$ the least depth guaranteeing this error for all valid trees has order $\Theta(\log\log|G|)$ as $|G| \to \infty$ (Theorem 7.1, Proposition 7.2, and Corollary 7.3).

The proof converts the bounded-rank diameter theorem of Pyber–Szabó [18] into a base-block contraction estimate and amplifies it through the tree. An elementary support bound gives the matching order of depth. Over prime fields outside a set of relative density zero among the primes, strong uniform expansion gives a constant gap for every generating involutive block of bounded depth. For the unobstructed classes this improves the exponential factor to e^{-c4^m} (Corollary 7.4). Over arbitrary finite fields, a second consequence of uniform expansion bounds the number of exceptional conjugacy classes of generating involutive configurations at each fixed depth by $O(q^\delta)$ for every $\delta > 0$ (Corollary 7.5).

The connection with the structural theory is the geometry of fixed spaces: once a common fixed vector is absent, their angle measures how strongly averaging over two subgroups contracts. For the exterior octahedral pairs with intersection of order six, all simultaneously conjugate to a natural Hamilton pair, this angle admits an asymptotically optimal uniform bound.

Theorem F (The natural octahedral pair)**.** *For primes $q \equiv \pm 3 \pmod 8$, $q \neq 3$, the Hamilton construction defines exterior subgroups $A_q, B_q \cong S_4$ of $G_q = \mathrm{PGL}_2(q)$ with $A_q \cap B_q \cong S_3$ and $\langle A_q, B_q\rangle = G_q$. Every pair A_1, A_2 of exterior octahedral subgroups of G_q with $|A_1 \cap A_2| = 6$ is simultaneously conjugate to (A_q, B_q), and its fixed-space projections satisfy*

$$\max_{\rho\in\widehat{G_q},\,\rho\neq\mathbf{1}} \|P^\rho_{A_1}P^\rho_{A_2}\| = c_q \leq \frac{\sqrt{3}}{2}, \qquad c_q \longrightarrow \frac{\sqrt{3}}{2},$$

where c_q is the common value attained by the natural pair (Proposition 8.2 and Theorem 8.5).

The compression identity transfers the Ramanujan bound of Davidoff–Sarnak–Valette [9, Theorem 4.2.2] from the norm-3 quaternionic walk to the angle operator. Alon–Boppana on the associated Hecke graphs gives optimality. Geometrically, $\theta_q = \arccos c_q$ is the smallest fixed-space angle over all non-trivial irreducibles: $\theta_q \geq \pi/6$ and $\theta_q \to \pi/6$. Combined with an internal S_4 estimate, the angle bound gives uniform contraction for the corresponding tree blocks (Corollary 8.7).

Comparison with freely chosen random products. Amir, Blachar, Ghosh and Vishne [1] study *mixability*: the existence of a finite random product that is exactly uniform, with its steps and Bernoulli probabilities freely chosen. Every S_n is mixable in that sense; Angel–Holroyd [2] construct exact shuffles using lazy adjacent transpositions with suitable probabilities. Our question instead fixes fair probabilities and a commutator pattern, and asks for contraction uniformly over all valid configurations. Among S_n with $n \geq 5$ and transposition leaves, only S_5 has such a level. The distinction involves both the allowed products and the quantifiers; the non-identity target is what makes the derived series govern the threshold. Section 9 also compares the construction with word maps, conjugacy-class walks and expansion.

Origin. The motivation comes from Barrington's width-5 branching programs [5]: what is the distribution of their group product when each expanded group instruction is independently replaced by the identity with probability $1/2$? The companion paper [23] proves target-independence and a quantitative mixing bound for the S_5 construction on inconsistent reads. Here we establish the general structural law and its quantitative consequences. For the conditioned law, Theorem 2.4 and Lemma 2.5 give target-independence and an additive conditioning error. The fixed-group mixing estimate uses the classical Diaconis–Shahshahani bound [10]. Additional consequences include product-quotient obstructions, comparison with a leaf Cayley walk, and determinant and congruence obstructions.

Reading guide. After the model and the conditioned branching-program law in Section 2, readers interested in the structural results can proceed to Sections 3 and 3.9 for Theorems A and B, and to Section 5.1 for Theorem C. Sections 4 to 6 give obstruction mechanisms, examples and the classification of Theorem D. The quantitative consequences and arithmetic application are developed in Sections 7 and 8. Section 10 distinguishes exact certificates from numerical checks, and Section 11 records the remaining problems.

2. Preliminaries

Let G be a finite group. A *permutation branching program* of length L over G is a sequence of instructions $((i_1, g_1^{(0)}, g_1^{(1)}), \dots)$; on input $x \in \{0,1\}^\ell$ it outputs $\prod_{j=1}^{L} g_j^{(x_{i_j})}$. For a point function the program is built from a fixed *template* (a read sequence and a fixed *leaf element* $\sigma_j \in G$ per leaf): leaf j outputs σ_j when its literal matches T and the identity e otherwise; the binary commutator tree of Barrington's construction then combines the leaves.

Output laws. For a labelled tree τ, drawing all expanded reads independently and uniformly defines its *unconditional* output law μ_τ. For the associated point-function

program we also write $P_T = \mu_\tau$: the fixed template is implicit, and the unconditional law does not depend on which branch matches T. Conditioning on inconsistent or consistent paths gives D_T or C_T, respectively.

Notation. Representations are unitary, with carrier space V_ρ and dimension d_ρ. We write $\widehat{G}$ for the irreducible representations and $\mathbf{1}$ for the trivial one. For a probability measure μ, put $\widehat{\mu}(\rho) = \sum_g \mu(g)\rho(g)$ and $M_\tau(\rho) = \widehat{\mu_\tau}(\rho)$. The notation $\|v\|$ denotes the Hilbert-space norm and $\|A\|$ the operator norm; $\|A\|_{\mathrm{F}}$ denotes the Frobenius norm. We use V_ρ^H for the H-fixed subspace and $\mathrm{Fix}_\rho(s) = V_\rho^{\langle s\rangle}$ for a single element. When ρ is fixed, we may write $V = V_\rho$ and $V^H = V_\rho^H$. For $w \in G$, its conjugacy class is denoted by w^G.

The independent-read commutator-tree model. The model has two structural features, recorded as Lemmas 2.1 and 2.2: the L bits are read *independently*, and Barrington's program combines leaves through a balanced *commutator tree*. A level-k target has 2^k configurable leaves; the expanded independent-read program has 4^k leaf occurrences because inverse copies are read independently with the inverse distribution. Write $\alpha(\tau)$ for the deterministic commutator target of τ and $H(\tau)$ for the subgroup generated by its leaf labels. The configuration is *valid* when $\alpha(\tau) \neq e$. For a conjugation-closed leaf set C, a subgroup is C-generated if it is generated by its intersection with C; the uniform strict-contraction level is the first level at which every valid configurable C-tree contracts in all non-trivial irreducible representations.

For a transitive action of G on Ω we write $V = \mathbb{C}[\Omega]_0 = \{f : \Omega \to \mathbb{C} \mid \sum_\omega f(\omega) = 0\}$ for the *augmentation representation*. Its fixed spaces detect the orbits of the leaf subgroup, as described in Proposition 3.4.

Lemma 2.1 (Independent-read factorisation)**.** *The operator $M_\tau(\rho)$ factors over the expanded leaf occurrences. A point-function leaf with equiprobable branches $\{e, \sigma\}$ contributes the* leaf operator

$$P_\sigma(\rho) = \tfrac{1}{2}\big(I + \rho(\sigma)\big),$$

and for an involution leaf ($\sigma^2 = e$) this is the orthogonal projection *onto* $\mathrm{Fix}_\rho(\sigma)$.

Proof. Independence makes the expectation defining $M_\tau(\rho)$ factor over the disjoint reads; a leaf reading one bit with branches e, σ of probability $\frac{1}{2}$ each contributes $\frac{1}{2}(\rho(e) + \rho(\sigma)) = \frac{1}{2}(I + \rho(\sigma))$. If $\sigma^2 = e$ then $\rho(\sigma)$ is a self-adjoint unitary involution, so P_σ is self-adjoint and $P_\sigma^2 = \frac{1}{4}(I+\rho(\sigma))^2 = \frac{1}{4}(2I+2\rho(\sigma)) = P_\sigma$. □

Lemma 2.2 (Recursive commutator-tree operator)**.** *Let $\tau = [\tau_1, \tau_2]$ be a node with its two child trees. Fix ρ and put $X = M_{\tau_1}(\rho)$ and $Y = M_{\tau_2}(\rho)$. Then*

$$M_\tau(\rho) = XYX^\dagger Y^\dagger.$$

For a single leaf s, the operator is $P_s = P_s(\rho)$. Thus a node with leaves s, t has operator $P_s P_t P_s^\dagger P_t^\dagger$, equal to $(P_s P_t)^2$ when both leaves are involutions. Expanding the recursion gives a product of leaf averages and their adjoints; involution leaf averages are orthogonal projections.

Proof. The four expanded sub-programs read disjoint positions. Their outputs a, b, a', b' are independent, with a', b' distributed as the inverse words a^{-1}, b^{-1}.

Unitarity gives $\mathbb{E}\rho(a') = X^\dagger$ and $\mathbb{E}\rho(b') = Y^\dagger$. Independence therefore yields

$$\mathbb{E}[\rho(a)\rho(b)\rho(a')\rho(b')] = XYX^\dagger Y^\dagger.$$

For involution leaves, $P_s^\dagger = P_s$ and $P_t^\dagger = P_t$ by Lemma 2.1. □

Remark 2.3 (Independent occurrences and word evaluation). The two distributions

$$\underbrace{XYX^{-1}Y^{-1}}_{\text{same } X,Y \text{ (correlated)}} \qquad \text{and} \qquad \underbrace{XY\,\tilde{X}^{-1}\tilde{Y}^{-1}}_{\tilde{X},\tilde{Y} \text{ independent copies of } X,Y}$$

encode different dependence between repeated occurrences. Ordinary word evaluation uses the same value of X in X and X^{-1}. The independent-read model uses four independent blocks, so its Fourier transform factors as in Lemma 2.2. This factorisation gives the fixed-vector criterion of Lemma 3.1.

Mixing bound. The Diaconis–Shahshahani lemma [10] gives

$$4\, d_{\mathrm{TV}}(\mu, U_G)^2 \le \sum_{\rho \neq \mathbf{1}} d_\rho \|\widehat{\mu}(\rho)\|_{\mathrm{F}}^2,$$

the sum over non-trivial irreducibles. Since $\|M\|_{\mathrm{F}}^2 \le d_\rho \|M\|^2$, controlling $\|M_\tau(\rho)\|$ for every non-trivial ρ controls $d_{\mathrm{TV}}(\mu_\tau, U_G)$. Every total-variation upper bound below may also be capped at 1.

2.1. **The conditioned branching-program law.** The structural and spectral results below concern the unconditional law μ_τ. The following reductions from [23, Theorem 3 and Lemma 1] transfer its mixing estimates to conditioned branching-program laws.

Theorem 2.4 (T-independence)**.** *Let* $\mathrm{BP}_T, \mathrm{BP}_{T'}$ *be point-function programs built from the same template (same read sequence, same leaf element* σ_j *at each leaf), differing only in which branch is the matching one at the literals where* $T_i \neq T_i'$*. Then* $D_T = D_{T'}$ *over any group* G*.*

Proof. Let $S = \{i : T_i \neq T_i'\}$ and define $F_S : \{0,1\}^L \to \{0,1\}^L$ flipping b_j exactly when the literal read at step j lies in S. Then (i) F_S is an involution; (ii) it flips all readings of each S-variable simultaneously, hence preserves the consistency/inconsistency of each variable individually; and (iii) at every step the swap of the matching branch ($T \leftrightarrow T'$) and the bit flip cancel, so the output is unchanged: $\varphi_{T'}(b) = \varphi_T(F_S(b))$. As F_S is a measure-preserving involution of the inconsistent paths, change of variables gives $D_{T'} = D_T$. □

The same bijection proves T-independence over every finite group and extends to programs with a fixed wildcard pattern.

Lemma 2.5 (Conditioning)**.** *Assume* $0 < p = \Pr[\textit{consistent}] < 1$*, so both conditional laws are defined. Then* $P_T = (1-p)D_T + p\,C_T$*, hence*

$$d_{\mathrm{TV}}(D_T, U_G) \;\le\; d_{\mathrm{TV}}(P_T, U_G) + p \cdot d_{\mathrm{TV}}(D_T, C_T) \;\le\; d_{\mathrm{TV}}(P_T, U_G) + p.$$

If every variable is read at least once then the consistent paths are in bijection with $\{0,1\}^\ell$*, so* $p = 2^\ell/2^L$*.*

Proof. The first identity is the law of total probability over the consistency event; the triangle inequality and $d_{\mathrm{TV}} \le 1$ give the bound. The count of consistent paths is the number of per-variable constant assignments. □

Any bound on $d_{\mathrm{TV}}(P_T, U_G)$ therefore gives a bound for D_T with additive loss $p = 2^{\ell - L}$. For $L = \ell^2$, this loss is $2^{-\Theta(\ell^2)}$. We now estimate the unconditional mixing term.

3. The mixing criterion

3.1. When does an operator fail to contract?

Lemma 3.1 (Geometric lemma)**.** *Fix a configuration τ with leaf set $S \subseteq G$ and let $H = \langle S \rangle$. For any unitary representation ρ on V,*

$$\|M_\tau(\rho)\| = 1 \quad \Longleftrightarrow \quad V^H \neq 0.$$

Proof. Each leaf operator $P_\sigma(\rho) = \frac{1}{2}(I + \rho(\sigma))$ is a contraction: $\rho(\sigma)$ is unitary with eigenvalues $e^{i\theta}$, so $P_\sigma(\rho)$ has eigenvalues $\frac{1}{2}(1+e^{i\theta})$ of modulus $|\cos(\theta/2)| \le 1$, equal to 1 only on the $+1$-eigenspace of $\rho(\sigma)$, i.e. on $\mathrm{Fix}_\rho(\sigma)$. By Lemma 2.2, $M_\tau(\rho)$ is a product $M_1 \cdots M_N$ in which each M_t is a leaf operator $P_\sigma(\rho)$ or its adjoint $P_\sigma(\rho)^\dagger = P_{\sigma^{-1}}(\rho)$ (some $\sigma \in S$); both are normal contractions whose norm-1 eigenspace is exactly $\mathrm{Fix}_\rho(\sigma) = \mathrm{Fix}_\rho(\sigma^{-1})$. If $\|M_1 \cdots M_N v\| = \|v\|$ for a unit vector v, put $v_N = v$ and $v_{t-1} = M_t v_t$. Since every M_t is a contraction and $\|v_0\| = \|v_N\|$, all inequalities $\|v_{t-1}\| \le \|v_t\|$ are equalities. Equality for M_t forces $v_t \in \mathrm{Fix}_\rho(\sigma_t)$, so $M_t v_t = v_t$ and $v_{t-1} = v_t$. Inducting gives $v_t = v$ for all t, hence v is fixed by every leaf, i.e. lies in $\bigcap_{\sigma \in S} \mathrm{Fix}_\rho(\sigma) = V^H$. Conversely any $v \in V^H$ is fixed by $M_\tau(\rho)$. As $\|M_\tau(\rho)\| \le 1$ always, the claim follows. □

Theorem 3.2 (Fourier fixed-vector criterion (Theorem B))**.** *The Fourier operator of a configuration τ contracts strictly in* every *non-trivial irreducible representation* if and only if its leaves generate G*:*

$$\|M_\tau(\rho)\| < 1 \ \ \forall \rho \neq \mathbf{1} \quad \Longleftrightarrow \quad H(\tau) = G.$$

Proof. Put $H = H(\tau)$. By Lemma 3.1 the left side fails for some non-trivial ρ iff $V_\rho^H \neq 0$ for some non-trivial ρ, i.e. iff $\mathrm{Ind}_H^G \mathbf{1} = \mathbb{C}[G/H]$ contains a non-trivial irreducible. Now $\mathbb{C}[G/H] = \bigoplus_\rho (\dim V_\rho^H)\, \rho$ has dimension $[G : H]$ and contains the trivial representation exactly once; it contains a non-trivial irreducible iff $[G : H] > 1$, i.e. iff $H \neq G$. Hence all non-trivial ρ contract iff $H = G$. □

Remark 3.3. Combined with submultiplicativity $\|ABA^\dagger B^\dagger\| \le \|A\|^2 \|B\|^2 \le \max\{\|A\|, \|B\|\}^4$ and the Diaconis–Shahshahani bound, Theorem 3.2 says: in a fixed finite instance, once every sub-block at some level k_0 has leaves generating G, every non-trivial Fourier block is strictly contractive; taking the maximum over the finitely many sub-blocks and irreducibles gives a gap < 1, and further levels then decay. For a family of programs, however, a quantitative total-variation bound also needs an explicit lower bound on this gap and on the determinant/congruence layers.

3.2. **The augmentation representation is only one test.**

Proposition 3.4 (Augmentation-representation specialisation)**.** *Assume G acts transitively on Ω. For a configuration τ and the augmentation representation $V = \mathbb{C}[\Omega]_0$, $\|M_\tau(V)\| = 1$ iff $H = H(\tau)$ is* intransitive *on Ω (has ≥ 2 orbits). Consequently the transitivity condition is* necessary but not sufficient *for strict contraction in every non-trivial irreducible: a proper* transitive *leaf subgroup contracts in V yet, by Theorem 3.2, fails to contract in some other irreducible.*

Proof. The space

$$V^H = \{f \in \mathbb{C}[\Omega]_0 : f \text{ is constant on } H\text{-orbits}\}$$

has dimension (number of H-orbits)-1, nonzero iff H is intransitive; apply Lemma 3.1. Intransitive $\Rightarrow$ $H \neq G$, but the converse fails, and any proper transitive H witnesses insufficiency. □

For S_5 acting on the 6 points of $\mathrm{PGL}_2(\mathbb{F}_5)$'s projective line, a scan of the 27420 reduced valid level-2 all-involution configurations (Section 10) finds the augmentation representation flags *none*, yet 4695 have leaves generating a proper transitive subgroup (3705 generating A_5, 990 an S_4-type subgroup of order 24); these are invisible to V but caught by the sign representation (for the A_5 ones) and others. Over S_5 in its natural 5-point action the two tests happen to coincide because the leaves there are transpositions, whose generated subgroups are Young subgroups; every proper such subgroup is intransitive.

3.3. **From the criterion to a solvability dichotomy.**

Lemma 3.5 (Validity is one-directional)**.** *The level-k commutator target of a configuration lies in $H^{(k)}$, the k-th derived subgroup of $H = \langle S \rangle$ (the leaf subgroup). In particular, if $H^{(k)} = \{e\}$ the target is trivial and the configuration is* invalid*.*

Proof. Induct on k. A level-0 target is a single leaf $\sigma \in S \subseteq H = H^{(0)}$. A level-$(k+1)$ target is a commutator $[A, B]$ of two level-k targets $A, B \in H^{(k)}$ (inductive hypothesis); hence $[A, B] \in [H^{(k)}, H^{(k)}] = H^{(k+1)}$. The converse fails: $H^{(k)} \neq \{e\}$ does not force a specific target to be non-trivial. □

By Theorem 3.2 a contraction failure (in some irreducible) at level k requires a *proper* leaf subgroup $H \neq G$ with $H^{(k)} \neq \{e\}$. To turn this into a dichotomy we fix a *conjugation-closed* leaf set C (a union of conjugacy classes of G); all our examples—transpositions in S_n, involutions in A_5 or PGL_2—are of this form. The key fact is that, ranging over such a set, the realisable targets generate the terms of the derived series.

Lemma 3.6 (Targets generate the derived series; cf. [12])**.** *Let $C \subseteq G$ be closed under conjugation by $H := \langle C \rangle$. Define the realisable level-k targets by $T_0 = C$ and $T_{k+1} = \{[a, b] : a, b \in T_k\}$, where a, b range independently (the left and right sub-blocks of a balanced commutator tree read disjoint inputs and may be configured independently). Then $\langle T_k \rangle = H^{(k)}$ for every $k \geq 0$.*

Proof. Induct on k; the base case is $\langle C\rangle = H = H^{(0)}$. Conjugation by $g \in H$ sends $[a,b] \mapsto [a^g, b^g]$ with $a^g, b^g \in T_k$, since C—and hence each T_k—is closed under H-conjugation; thus T_{k+1} is H-conjugation-closed and $\langle T_{k+1}\rangle$ is normal in H. Write $M = H^{(k)} = \langle T_k\rangle$. Each $[a,b]$ $(a,b \in T_k \subseteq M)$ lies in $[M,M] = H^{(k+1)}$, so $\langle T_{k+1}\rangle \subseteq H^{(k+1)}$. Conversely, $[M,M]$ is the normal closure in M of $\{[s,t] : s,t \in T_k\}$ (in M/N, with N that normal closure, the images of the generating set T_k commute, so M/N is abelian); as $\langle T_{k+1}\rangle$ is normal in M and contains those commutators, it contains $[M,M] = H^{(k+1)}$. Hence equality. □

Remark 3.7 (Configurable leaf sets and rigid templates). In the configurable model, the two sub-blocks range independently over the conjugation-closed leaf set. Their targets generate $H^{(k)}$ by Lemma 3.6, giving the exact threshold in Theorem 3.9. A rigid template specifies a leaf at each position; its realised targets form a restricted subset of the configurable targets. The same value D then gives an upper bound. Section 5.1 studies attainable rigid levels and persistence of valid binary continuations.

Definition 3.8 (C-generated subgroup)**.** A subgroup $H \leq G$ is *C-generated* if $H = \langle H \cap C\rangle$. Every leaf subgroup $L = \langle S\rangle$ with $S \subseteq C$ is C-generated; and, C being conjugation-closed in G, $L \cap C$ is closed under conjugation by L.

Theorem 3.9 (Solvability dichotomy and exact level (Theorem A))**.** *Let G be a finite* non-solvable *group and $C \subseteq G$ a conjugation-closed leaf set with $\langle C\rangle = G$, in the* configurable *model where the two sub-blocks of each node may be configured independently (Remark 3.7; for a single rigid template the level D below is only an upper bound).* (A solvable ambient is degenerate: no valid configuration of large level exists, its target lying in $G^{(d)} = \{e\}$.)

(1) ***(Obstruction.)*** *If some proper C-generated subgroup is non-solvable, then for every level k a valid configuration τ has $\|M_\tau(\rho)\| = 1$ for some non-trivial ρ: the leaf set has* no uniform strict-contraction level.

(2) ***(Sharp level.)*** *If every proper C-generated subgroup is solvable, put $D = \max\{\mathrm{dl}(H) : H < G,\ H \text{ is } C\text{-generated}\}$. Then every valid configuration of level $\geq D$ has leaf subgroup G and contracts strictly in all non-trivial irreducibles, while some valid configuration of level $D-1$ does not. The level of strict contraction is exactly D.*

Proof. (1) Let $H < G$ be proper, C-generated and non-solvable; its derived series stabilises at a non-trivial perfect core, so $H^{(k)} \neq \{e\}$ for all k. By Lemma 3.6 inside H (with the H-conjugation-closed set $H \cap C$, which generates H by Definition 3.8), the level-k targets with leaves in $H \cap C$ generate $H^{(k)} \neq \{e\}$, so a valid level-k configuration τ with leaves in H exists. Its leaf subgroup is $\leq H < G$, so by Theorem 3.2 some non-trivial ρ has $V_\rho^{H(\tau)} \neq 0$, whence $\|M_\tau(\rho)\| = 1$ (Lemma 3.1). *(2)* If a valid level-k configuration $(k \geq D)$ had leaf subgroup $L \neq G$, then L is proper, C-generated and, by hypothesis, solvable with $\mathrm{dl}(L) \leq D \leq k$, so its target lies in $L^{(k)} \subseteq L^{(D)} = \{e\}$ (Lemma 3.5)—contradicting validity. Thus $L = G$ and it contracts strictly (Theorem 3.2). For exactness, take proper C-generated H with $\mathrm{dl}(H) = D$; by Lemma 3.6 inside H the level-$(D-1)$ targets generate $H^{(D-1)} \neq \{e\}$,

giving a valid level-$(D-1)$ configuration with leaf subgroup $\leq H < G$ that fails to contract (Lemma 3.1, Theorem 3.2). □

The contraction level is therefore exactly the largest derived length among proper C-generated subgroups, and yields a quantitative bound for each fixed group. The value D is also the first level at which uniform contraction holds: if a proper C-generated H has $\mathrm{dl}(H) = D$, then $H^{(k)} \neq \{e\}$ for every $k < D$, and Lemma 3.6 inside H supplies a valid configuration with leaf subgroup contained in H at each of these levels, so strict contraction fails at every level below D.

Proposition 3.10 (Quantitative bound, fixed group)**.** *In case (2) of Theorem 3.9, let $\lambda = \lambda_{G,C} < 1$ be the maximum of $\|M_\tau(\rho)\|$ over non-trivial irreducibles ρ and valid level-D configurations τ. For every valid configuration τ of level $m \geq D$,*

$$d_{\mathrm{TV}}(\mu_\tau, U_G) \leq \min\left\{1, \tfrac{1}{2}\sqrt{|G|-1}\,\lambda^{4^{m-D}}\right\}.$$

For the associated conditioned law D_T, add p from Lemma 2.5 to this bound.

Proof. Every sub-block of a valid configuration is valid: were a sub-block target e, each ancestor commutator $[e, \cdot] = e$ would vanish, forcing the root target to be e. Hence each level-D sub-block has leaf subgroup G and operator norm $\leq \lambda$ (case (2)), and submultiplicativity gives $\|M_\tau(\rho)\| \leq \lambda^{4^{m-D}}$. With $\|M\|_{\mathrm{F}}^2 \leq d_\rho \|M\|^2$ and $\sum_\rho d_\rho^2 = |G|$, Diaconis–Shahshahani gives $4d_{\mathrm{TV}}(\mu_\tau, U_G)^2 \leq (|G|-1)\lambda^{2\cdot 4^{m-D}}$; the D_T bound is Lemma 2.5. For $G = S_5$ the prefactor $\frac{1}{2}\sqrt{119} \approx 5.45$ recovers the companion paper's bound. Section 7 controls the gap as the group varies and derives uniform family rates. □

3.4. **Functoriality and localisation of obstructions.** The dichotomy is monotone under quotients: any obstruction visible in a quotient is already an obstruction upstairs. Thus, to mix, G must mix through every non-solvable quotient; solvable quotients are the degenerate case already noted in Theorem 3.9.

Proposition 3.11 (Quotients)**.** *Let $N \trianglelefteq G$ and $\bar{C} = \pi(C) \subseteq G/N$ its image under $\pi : G \to G/N$ (again conjugation-closed). If (G, C) is in the mixing case of Theorem 3.9, then every proper $\bar{C}$-generated subgroup of G/N is solvable. In particular, if G/N is non-solvable, then $(G/N, \bar{C})$ is also in the mixing case, and the level cannot rise:*

$$D(G/N, \bar{C}) \leq D(G, C).$$

Equivalently, an obstruction in a quotient lifts to one in G.

Proof. Let $\bar{K} < G/N$ be proper and $\bar{C}$-generated, and set $L = \langle C \cap \pi^{-1}(\bar{K})\rangle$. Then L is C-generated, and since $\pi\big(C \cap \pi^{-1}(\bar{K})\big) = \bar{C} \cap \bar{K}$ we have $\pi(L) = \langle \bar{C} \cap \bar{K}\rangle = \bar{K}$. Hence $L < G$ is proper and $\mathrm{dl}(L) \geq \mathrm{dl}(\pi(L)) = \mathrm{dl}(\bar{K})$. So a non-solvable $\bar{K}$ forces a non-solvable L (the obstruction lifts); and if every proper C-generated subgroup of G is solvable, so is L, whence $\bar{K}$ is solvable with $\mathrm{dl}(\bar{K}) \leq \mathrm{dl}(L) \leq D(G, C)$. Maximising over $\bar{K}$ gives the level bound when G/N is non-solvable. □

Corollary 3.12 (An obstruction from product quotients)**.** *Suppose a finite group G has a quotient $A \times B$, with A non-solvable and B non-cyclic. Every conjugation-closed generating leaf set of G then has a permanent obstruction in the configurable model.*

Proof. First suppose $G = A \times B$ and let C generate G. The first projection of C generates A, so some $(a, b) \in C$ has $a \notin O_\infty(A)$. The normal closure of a in A is non-solvable: otherwise it would be contained in the solvable radical. Conjugation-closure implies

$$S = \{(xax^{-1}, b) : x \in A\} \subseteq C.$$

The group $H = \langle S \rangle$ projects onto that non-solvable normal closure, while $H \leq A \times \langle b \rangle < A \times B$ since B is non-cyclic. Thus H is a proper non-solvable C-generated subgroup. For a quotient of G, apply this argument to the image leaf set and then lift the obstruction by Proposition 3.11. □

The non-cyclic hypothesis cannot simply be omitted. For example, let $G = A_5 \times C_2$ and take $C = \{(t, z) : t \in A_5 \text{ an involution}\}$, where z generates C_2. This class generates G: a subgroup projecting onto A_5 with trivial kernel would be the graph of a homomorphism $A_5 \to C_2$, which cannot contain such leaves. Similarly, a proper C-generated subgroup cannot project onto all of A_5 and is therefore contained in $K \times C_2$ for a proper involution-generated $K < A_5$. These K have derived length at most 2 by Corollary 5.2; the bound is attained by lifting the involutions of an S_3 along its sign character. Hence this class has exact level 2. In contrast, $A_5 \times (C_2 \times C_2)$ admits no mixing conjugation-closed generating class.

Proposition 3.13 (Perfect-core localisation)**.** *For a finite group write $H^{(\infty)} = \bigcap_i H^{(i)}$ for the terminal (perfect) term of the derived series, so H is solvable iff $H^{(\infty)} = 1$. Let $N \trianglelefteq G$, $\pi\colon G \to G/N$, and let $H < G$ be a proper non-solvable C-generated subgroup (an obstruction in Theorem 3.9) with $HN \neq G$. Then $\pi(H)$ is a proper $\bar{C}$-generated subgroup, and exactly one holds:*

(1) $\pi(H)$ is non-solvable — the obstruction descends*, $\pi(H)$ being an obstruction for $(G/N, \bar{C})$; or*

(2) $\pi(H)$ is solvable — the perfect core falls into the kernel, $H^{(\infty)} \leq H \cap N$.

Proof. $HN \neq G$ gives $\pi(H) = HN/N < G/N$ proper, and $H = \langle H \cap C \rangle$ gives $\pi(H) = \langle \pi(H \cap C) \rangle$, hence $\bar{C}$-generated. Since $\pi(H^{(i)}) = \pi(H)^{(i)}$ and a finite derived series stabilises, $\pi(H^{(\infty)}) = \pi(H)^{(\infty)}$. If $\pi(H)$ is solvable then $\pi(H)^{(\infty)} = 1$, so $H^{(\infty)} \leq \ker \pi \cap H = H \cap N$; otherwise $\pi(H)$ is non-solvable. (Should a leaf lie in N, $e \in \bar{C}$; discarding it changes no generated subgroup.) □

Remark 3.14 (Schreier generators for kernel obstructions). In case (ii), the kernel $K = H \cap N$ may contain no leaves. For example, for $G = S_5 \times C_2$, $N = A_5 \times 1$, leaves the transpositions of S_5 and a generator of C_2, the obstruction $H = S_5 \times 1$ has $H^{(\infty)} = A_5 \times 1 \leq N$ while $C \cap N = \varnothing$. Reidemeister–Schreier supplies generators: for $H = \langle S \rangle$ and $K = H \cap N$, choosing representatives $t \in H$ for the cosets $K\backslash H \simeq \pi(H)$ gives generators

$$t\, s\, \overline{ts}^{-1} \qquad (s \in S),$$

where $\overline{ts}$ denotes the chosen representative of the coset Kts. If the representatives are chosen by shortest words in the image generating set $\pi(S)$, these Schreier words have length at most $2\,\mathrm{diam}(\pi(H), \pi(S)) + 1$ in the original leaves. The kernel inherits the leaves' *lifting errors*, providing a mechanism for congruence-layer obstructions in $\mathrm{GL}_n(\mathbb{Z}/p^r)$ (Section 4).

Proposition 3.15 (Clean reduction over a Frattini solvable kernel)**.** *Let $N \trianglelefteq G$ be solvable with $N \le \Phi(G)$, the Frattini subgroup. Then no proper subgroup supplements N ($HN = G \Rightarrow H = G$), and in the non-degenerate regime (G non-solvable and $\langle C \rangle = G$, hence G/N non-solvable and $\langle \bar{C} \rangle = G/N$)*

$$(G, C) \text{ mixes} \iff (G/N, \bar{C}) \text{ mixes},$$

$$\text{in the mixing case:} \quad D(G/N, \bar{C}) \le D(G, C) \le D(G/N, \bar{C}) + \mathrm{dl}(N).$$

Proof. If $H < G$ had $HN = G$, then $H \le M$ for some maximal $M \supseteq \Phi(G) \supseteq N$, whence $G = HN \le M < G$; so $HN = G$ forces $H = G$, and every proper C-generated H has $\pi(H)$ proper. If H is a non-solvable obstruction then $H^{(\infty)} \ne 1$, while $H^{(\infty)} \le N$ would make it a perfect subgroup of the solvable N and force $H^{(\infty)} = 1$; so case (ii) of Proposition 3.13 is impossible and $\pi(H)$ is a non-solvable obstruction in G/N. With Proposition 3.11 this gives the equivalence. For the level, a proper solvable C-generated H has $\mathrm{dl}(H) \le \mathrm{dl}(\pi(H)) + \mathrm{dl}(H \cap N) \le D(G/N, \bar{C}) + \mathrm{dl}(N)$ (with $\mathrm{dl}(1) = 0$); the left inequality is Proposition 3.11. □

Remark 3.16 (Supplements to the kernel). A proper supplement $H < G$ with $HN = G$ maps onto the whole quotient. When G/N is non-solvable, so is H, and detecting this obstruction requires information about supplements. Corollary 3.12 gives one such obstruction for product quotients.

3.5. **Stability of the criterion.** The fixed-vector criterion has a quantitative form: a vector that loses little norm is close to the fixed space of the leaf subgroup. The estimate applies to leaves of arbitrary order.

Proposition 3.17 (Stability / quantitative gap)**.** *Let $M_j = (I + \rho(s_j))/2$, $T = M_N \cdots M_1$, and $H = \langle s_1, \dots, s_N \rangle \ne \{e\}$, where ρ is unitary on V and the s_j may have arbitrary orders. Let L be the diameter of H in $\{s_1, \dots, s_N\}^{\pm 1}$. For a unit vector v, put $E = 1 - \|Tv\|^2$. Then*

$$\|(I - M_j)v\| \le \sqrt{NE}, \qquad \mathrm{dist}(v, V^H) \le 2L\sqrt{NE}.$$

In particular, if $V^H = 0$, then

$$\|T\|^2 \le 1 - \frac{1}{4NL^2}, \qquad \|T\| \le 1 - \frac{1}{8NL^2}. \tag{3.1}$$

Proof. For a unitary U and $M = (I + U)/2$, the parallelogram identity gives $\|Mx\|^2 + \|(I - M)x\|^2 = \|x\|^2$, and $\|I - M\| \le 1$. Set $w_0 = v$, $w_j = M_j w_{j-1}$ and $e_j = w_{j-1} - w_j$. Thus $\sum_j \|e_j\|^2 = E$ and

$$\|(I - M_j)v\| \le \|e_j\| + \|v - w_{j-1}\| \le \sum_{i \le j} \|e_i\| \le \sqrt{NE}.$$

Consequently $\|(I-\rho(s_j))v\| \le 2\sqrt{NE}$; the same holds for s_j^{-1}. Telescoping a word of length at most L gives $\|(I-\rho(h))v\| \le 2L\sqrt{NE}$ for every $h \in H$. Averaging over H proves the distance bound. If $V^H = 0$, this yields $1 \le 2L\sqrt{NE}$ for every unit v, proving the first inequality in (3.1); use $\sqrt{1-x} \le 1 - x/2$ for the second. □

Remark 3.18. For generating leaves in a non-trivial irreducible, Proposition 3.17 gives an explicit gap in terms of their Cayley diameter. With $N = 4^d$ at level d, this estimate yields the family rates in Section 7.

Proposition 3.19 (Comparison with the leaf Cayley walk)**.** *Let $s_1, \dots, s_N$ be involutions in a finite group G, listed with multiplicity and $N \ge 1$. On $\ell_0^2(G)$ let R be the regular representation and put*

$$P_i = \tfrac{1}{2}(I + R(s_i)), \qquad T = P_N \cdots P_1, \qquad A = \frac{1}{N}\sum_{i=1}^{N} P_i, \qquad \delta = 1 - \|A\|.$$

Then

$$\frac{2\delta}{N+1} \;\le\; 1 - \|T\|^2 \;\le\; \min\{1, 4N\delta\}.$$

Here A is the positive semidefinite transition operator of the lazy Cayley walk with the indicated multiplicities. The assertions include $\delta = 0$, which occurs exactly when the leaves generate a proper subgroup.

Proof. For a unit vector v write $w_0 = v$, $w_i = P_i w_{i-1}$ and $e_i = (I - P_i)w_{i-1}$. Orthogonality gives $\sum_i \|e_i\|^2 = 1 - \|Tv\|^2$. Also

$$\|(I-P_i)v\| \le \sum_{j \le i} \|e_j\|, \qquad \sum_i \|(I-P_i)v\|^2 \le \frac{N(N+1)}{2} \sum_j \|e_j\|^2.$$

Since $I - A \ge \delta I$, the left sum is at least $N\delta$. Choose v attaining $\|T\|$ to obtain the lower bound. For the upper bound choose a unit eigenvector v of $I - A$ with eigenvalue δ. Gao's projection union bound, in the form [3, Lemma 2], gives

$$\|Tv\|^2 \ge 1 - 4\sum_i \|(I-P_i)v\|^2 = 1 - 4N\delta.$$

The additional bound by 1 follows from $\|T\|^2 \ge 0$. □

Corollary 3.20 (Bounded-depth uniformity)**.** *For a family of involutive independent-read blocks with uniformly bounded expanded length N, their operator norms on $\ell_0^2(G)$ are uniformly bounded below 1 if and only if the associated lazy Cayley walks have uniformly positive spectral gaps. In a level-k block $N = 4^k$; the walk uses the expanded occurrences, equivalently the 2^k leaf positions with their multiplicities. Thus the assertion applies to uniformly bounded depths, and asserts no such equivalence with uniform constants when depths are unbounded.*

Proof. Both implications follow from Proposition 3.19; the regular representation contains every non-trivial irreducible, so its norm on $\ell_0^2(G)$ is the maximum of their norms. □

3.6. **Averaging leaves.** The fixed-vector criterion extends to general lazy averages. For a probability distribution μ on G with $e \in \operatorname{supp}\mu$, put $P_\mu(\rho) = \mathbb{E}_{g\sim\mu}[\rho(g)] = \sum_g \mu(g)\rho(g)$.

Proposition 3.21 (Averaging leaves)**.** *For lazy μ, $\|P_\mu\| \le 1$, and $\|P_\mu v\| = \|v\|$ iff $v \in V_\rho^{H_\mu}$ with $H_\mu = \langle \operatorname{supp}\mu\rangle$ (and then $P_\mu v = v$). Hence Lemma 3.1 and Theorem 3.2 hold verbatim with H_μ in place of $\langle\sigma\rangle$: averaging leaves contract in every non-trivial irreducible iff $\langle\bigcup_{leaves} H_\mu\rangle = G$. The point-function leaf $\mu = \operatorname{unif}\{e,\sigma\}$ is the case $H_\mu = \langle\sigma\rangle$.*

Proof. $\|\sum_g \mu(g)\rho(g)v\| \le \sum_g \mu(g)\|\rho(g)v\| = \|v\|$, with equality iff the equal-norm vectors $\rho(g)v$ ($g \in \operatorname{supp}\mu$) coincide; as $e \in \operatorname{supp}\mu$ they then all equal $\rho(e)v = v$, i.e. $v \in V_\rho^{H_\mu}$, and $P_\mu v = \sum_g \mu(g)v = v$. The peeling of Lemma 3.1 is unchanged. □

Remark 3.22. For these averaging operators, the generated supports determine the common fixed space. The target and validity conditions in Theorem 3.9 apply when the underlying leaf labels range over a configurable conjugation-closed class.

3.7. **Subgroup averages and finite Hecke boundary operators.** Subgroup averages connect fixed spaces to the angles used in Section 8 through finite Hecke operators. Fix a unitary representation ρ on V. In this subsection we write $P_H = P_H^\rho$ for the projection onto V^H.

Lemma 3.23 (Subgroup averages are fixed-space projections)**.** *Let $H \le G$ and set*

$$e_H = \frac{1}{|H|}\sum_{h\in H} h \in \mathbb{C}[G], \qquad P_H = \rho(e_H) = \frac{1}{|H|}\sum_{h\in H}\rho(h).$$

Then P_H is the orthogonal projection onto V^H.

Proof. The average is self-adjoint and idempotent. Its image is H-fixed by reindexing the sum, and it acts as the identity on every H-fixed vector. □

Proposition 3.24 (Subgroup averaging and finite Hecke compression)**.** *Let $H \le G$, let $g \in G$, and let U_H be the uniform probability measure on H. Then*

$$\widehat{U_H * \delta_g * U_H}(\rho) = P_H\rho(g)P_H.$$

The measure $U_H * \delta_g * U_H$ depends only on the double coset HgH, and the corresponding group-algebra element belongs to the finite Hecke corner $e_H\mathbb{C}[G]e_H$.

Proof. Convolution becomes multiplication under ρ. For $h_1, h_2 \in H$, $e_H h_1 g h_2 e_H = e_H g e_H$, proving the double-coset assertion. □

Corollary 3.25 (Double cosets as angles between fixed spaces)**.** *Let $K = gHg^{-1}$. Then*

$$P_K = \rho(g)P_H\rho(g)^{-1}, \qquad P_H\rho(g)P_H = P_H P_K \rho(g),$$

and therefore

$$\|P_H\rho(g)P_H\| = \|P_H P_K\|.$$

Thus the norm of the double-coset compression is exactly the cosine of the smallest principal angle between the fixed spaces V^H and V^K, with cosine defined as 0 *when either space is zero.*

Proof. Conjugate the average over H, then use $P_K\rho(g) = \rho(g)P_H$. Right multiplication by the unitary $\rho(g)$ preserves the norm. □

If $\langle H, K\rangle = G$ and ρ is non-trivial irreducible, then $V^H \cap V^K = V^G = 0$. The angle in Corollary 3.25 is therefore the Friedrichs angle. This is the angle controlling classical alternation of two orthogonal projections; see [4] for that formula and its distinction from products involving more than two subspaces. The tree estimates here use their stated product bounds, and Lemma 8.6 also allows blocks that only approximate fixed-space projections.

Lemma 3.26 (Local-to-boundary stability)**.** *Let $H, K \le G$, let $g \in G$, and let ν_H, ν_K be probability measures supported on H, K. Put $X = \widehat{\nu_H}(\rho)$ and $Y = \widehat{\nu_K}(\rho)$. If*

$$\|X - P_H\| \le \varepsilon_H, \qquad \|Y - P_K\| \le \varepsilon_K,$$

then

$$\|X\rho(g)Y - P_H\rho(g)P_K\| \le \varepsilon_H + \varepsilon_K.$$

Proof. Insert $P_H\rho(g)Y$ between the two terms. The triangle inequality and the fact that all other factors are contractions give $\varepsilon_H + \varepsilon_K$. □

Remark 3.27. An internal contraction estimate controls the error in replacing a block by a subgroup average; the angle then controls the interaction of two blocks. Lemma 8.6 and Theorem 8.5 implement this for the octahedral pairs with intersection of order six.

3.8. **Effectivity.**

Proposition 3.28 (Decidability)**.** *Given an effective finite group model in which subgroups can be enumerated (for example, a multiplication table, or permutation groups via standard algorithms) and a conjugation-closed leaf set C, it is decidable whether (G, C) is in the mixing case of Theorem 3.9; when it is, the exact level D is computable, and otherwise a non-solvable obstructing subgroup is exhibited (or the ambient/input is reported as degenerate, e.g. $\langle C\rangle \neq G$ or G solvable).*

Proof. First check whether $\langle C\rangle = G$ and whether the ambient is non-solvable. Then enumerate the subgroups $H < G$, retain exactly the C-generated ones ($H = \langle H \cap C\rangle$), and test each for solvability by its derived series. Taking the maximum derived length among the solvable witnesses returns D; a non-solvable one is the obstruction in Theorem 3.9. □

Remark 3.29. The cost of the procedure is governed by the number of subgroups enumerated.

The worked instances—S_n, A_5, $\mathrm{PGL}_2(\mathbb{F}_q)$—are collected in Section 5; the matrix groups $\mathrm{GL}_n(\mathbb{F}_q)$ reduce to a subgroup question (Section 4).

3.9. **The analytic–algebraic correspondence.** The expanded independent-read operator is a product of lazy averages. Its norm-one condition depends on the subgroup generated by the leaves, as in Lemma 3.1 and Theorem 3.2. The derived series enters through the additional requirement that the folded commutator target be non-identity. The normal-subset word-value lemma [12, Lemma 2.1(ii)] identifies the subgroup generated by the realisable depth-k targets with $H^{(k)}$. Combining these two facts gives the exact obstruction and level of Theorem 3.9.

The quantitative correspondence is recorded in Proposition 3.19: for involution leaves it compares the block gap with that of the associated lazy Cayley walk. Table 1 collects the connections used throughout the paper.

Tree quantity	Correspondence
Leaf average	A lazy group step; an orthogonal fixed-space projection for an involution.
Strict contraction	Generation by the leaves, when tested in every non-trivial irreducible (Theorem B).
First uniform level	Maximal proper leaf-subgroup derived length, in the unobstructed configurable case (Theorem A).
Uniform block gap	Expansion of the associated lazy Cayley walks, for involutions at bounded depth (Corollary 3.20).
Two subgroup averages	Fixed-space angle; for conjugate subgroups, the norm of a finite Hecke compression (Corollary 3.25).

TABLE 1. Connections used in the structural and quantitative arguments.

4. THE THREE LAYERS AND MATRIX OBSTRUCTIONS

For $G = \mathrm{GL}_n(\mathbb{F}_q)$ the determinant $\det : G \twoheadrightarrow \mathbb{F}_q^\times$ is an abelian quotient with kernel SL_n. Three sources of obstructions are useful in these matrix families: subspace stabilisers, the determinant quotient, and congruence kernels over $\mathbb{Z}_{p^k}$. We examine each through the leaf-subgroup criterion.

(a) The shape, and matrix obstructions. The non-abelian content is Theorem 3.2 and Theorem 3.9. For the relevant matrix families, parabolics (subspace stabilisers) give an explicit family of obstructions. The stabiliser of an r-subspace of $\mathbb{F}_q^n$ has Levi factor $\mathrm{GL}_r \times \mathrm{GL}_{n-r}$, which is non-solvable as soon as one factor is. A leaf set whose elements can remain inside such a parabolic may therefore contain a proper non-solvable C-generated subgroup.

Proposition 4.1 (Parabolic obstruction in the matrix regime)**.** *Let $P < \mathrm{GL}_n(\mathbb{F}_q)$ be a proper parabolic and let $H_P = \langle P \cap C \rangle$. If the image of H_P in a Levi factor of P is non-solvable, then H_P is a proper non-solvable C-generated subgroup. Consequently the leaf set has a permanent obstruction in the sense of Theorem 3.9(1). Thus parabolics provide explicit structural obstructions whenever this Levi-image condition can be verified. By contrast, $\mathrm{GL}_2(\mathbb{F}_q)$ has only the Borel as a proper parabolic, and the Borel is solvable; $\mathrm{GL}_3(\mathbb{F}_2)$ has Levi $\mathrm{GL}_2(\mathbb{F}_2) \cong S_3$ (solvable), so this particular parabolic obstruction does not appear in these small cases.*

Parabolics furnish an explicit infinite family of obstructions; the remaining maximal subgroups reduce to a Dickson/Aschbacher generation question (Section 11).

Proposition 4.2 (Transvection leaves)**.** *Let $n \geq 3$, let $G = \mathrm{SL}_n(q)$, and let $\mathcal{T}$ be its class of transvections. The exact configurable contraction level is* 3 *for* $(n, q) = (3, 2)$ *and* 4 *for* $(n, q) = (3, 3)$*. In every other case there is a permanent obstruction.*

Proof. Transvections generate $\mathrm{SL}_n(q)$. For $q \geq 4$ the upper-left $\mathrm{SL}_2(q)$ is a proper non-solvable $\mathcal{T}$-generated subgroup. For $q \in \{2, 3\}$ and $n \geq 4$, use the upper-left $\mathrm{SL}_3(q)$ instead. Apply Theorem 3.9(1). In $\mathrm{SL}_3(2)$ transvections are precisely the involutions, and Remark 5.4 gives level 3.

For $\mathrm{SL}_3(3) = \mathrm{PSL}_3(3)$ use the maximal-subgroup list in Proposition 6.4. In a point parabolic $P = 3^2{:}\,\mathrm{GL}_2(3)$, a transvection acts trivially on the invariant line, so its Levi image lies in $\mathrm{SL}_2(3)$. The radical $V \cong \mathbb{F}_3^2$ and the elementary matrices in this Levi factor are transvections; hence $\langle P \cap \mathcal{T} \rangle = J = V \rtimes \mathrm{SL}_2(3)$. Since $\mathrm{SL}_2(3)' = Q_8$, $Q_8' = \langle -I \rangle$, and $-I$ acts as negation on V, the derived series is

$$J \;>\; V \rtimes Q_8 \;>\; V \rtimes \langle -I \rangle \;>\; V \;>\; 1,$$

with orders $216, 72, 18, 9, 1$. The dual parabolic has the same property. The other maximals, S_4 and 13:3, have derived lengths 3 and 2. Every proper $\mathcal{T}$-generated subgroup therefore has derived length at most 4, and J attains 4. Apply Theorem 3.9(2). □

(b) The determinant layer. Apart from the small exceptions to the commutator-subgroup statement (for instance $\mathrm{GL}_2(\mathbb{F}_2) \cong S_3$), the one-dimensional representations of $\mathrm{GL}_n(\mathbb{F}_q)$ are $\chi \circ \det$ for χ a character of the cyclic group $\mathbb{F}_q^\times \cong \mathbb{Z}_{q-1}$. The output determinant is a lazy random walk on $\mathbb{Z}_{q-1}$ with steps $\{s_j = \mathrm{dlog} \det \sigma_j\}$.

Proposition 4.3 (Determinant layer)**.** (Strict contraction.) *The determinant layer has no non-trivial character of modulus one iff* $\{\det \sigma_j\}$ *generate* $\mathbb{F}_q^\times$. (Fourier coefficient.) *For the character χ_t after L leaves the complex coefficient is* $\prod_j (1 + \chi_t(\det \sigma_j))/2$*, and its modulus is* $\prod_j |\cos(\pi t s_j/(q-1))|$*. Here L counts expanded occurrences, including inverse copies. No rate follows from generation alone without specifying the multiset of steps and the tolerance. For the separate lazy walk repeatedly using one generator of a cyclic group of order $n \geq 3$, mixing time at fixed tolerance in* $(0, 1/2)$ *is* $\Theta(n^2)$. (Leaves.) *An involution σ has* $\det \sigma \in \{\pm 1\}$*; thus for* $q - 1 > 2$ *involution leaves trap* det *in a proper subgroup, and full* $\mathrm{GL}_n(\mathbb{F}_q)$ *mixing* requires non-involution leaves *whose determinants generate* $\mathbb{F}_q^\times$. (In PGL_n, the determinant class lies in $\mathbb{F}_q^\times/(\mathbb{F}_q^\times)^n$, of order $\gcd(n, q-1)$; this is the square-class quotient only in the essentially $n = 2$ case.)

The displayed Fourier coefficients quantify the determinant contribution to mixing for the prescribed leaf multiset.

(c) The composite-modulus layer. For $q = p^k$, $1 \to K \to \mathrm{GL}_n(\mathbb{Z}/p^k) \to \mathrm{GL}_n(\mathbb{F}_p) \to 1$ with $K = 1 + pM_n(\mathbb{Z}/p^k\mathbb{Z})$ a nilpotent p-group. Valid trees at arbitrary depth require a non-solvable ambient group, and $\mathrm{GL}_n(\mathbb{Z}/p^k)$ is non-solvable iff $\mathrm{GL}_n(\mathbb{F}_p)$ is. Since $\mathrm{GL}_2(\mathbb{F}_2) \cong S_3$ and $\mathrm{GL}_2(\mathbb{F}_3)$ are solvable, $\mathrm{GL}_2(\mathbb{Z}_4)$

and $\mathrm{GL}_2(\mathbb{Z}_9)$ are solvable: indeed $\mathrm{GL}_2(\mathbb{Z}_4)$ has derived series $96 \to 24 \to 4 \to 1$, so every level-3 target is trivial. For $k = 2$, $K \cong M_n(\mathbb{F}_p)^+$ is abelian. Clifford theory then organises the irreducibles above non-trivial K-characters by stabilisers, orbits, and induction. The first non-degenerate 2×2 case is $\mathrm{GL}_2(\mathbb{Z}_{25})$, of order $3 \cdot 10^5$. For generating conjugacy classes, Serre's lifting lemma gives a direct reduction.

Proposition 4.4 (Congruence lifting with generating determinants)**.** *Let $p \geq 5$ be prime, $r \geq 2$, $G = \mathrm{GL}_2(\mathbb{Z}/p^r)$, and $K = \ker(G \to \mathrm{GL}_2(\mathbb{F}_p))$. Let C be conjugation-closed, generate G, and have the property that* det *c generates $(\mathbb{Z}/p^r)^\times$ for every $c \in C$. Then (G, C) has a uniform contraction level if and only if $(\mathrm{GL}_2(\mathbb{F}_p), \bar{C})$ does. In this case their exact levels satisfy*

$$\bar{D} \leq D \leq \bar{D} + \mathrm{dl}(K).$$

In particular, $\bar{D} \leq D \leq \bar{D} + 1$ for $r = 2$. The determinant hypothesis holds for every single conjugacy class that generates G.

Proof. Every non-trivial C-generated subgroup H has full determinant image. If its reduction is $\mathrm{GL}_2(\mathbb{F}_p)$, Serre's lifting lemma [16, Lemma 3.15], applied to the closed preimage of H in $\mathrm{GL}_2(\mathbb{Z}_p)$, implies that H contains $\mathrm{SL}_2(\mathbb{Z}/p^r)$. The full determinant image then gives $H = G$. Consequently every proper C-generated subgroup has proper reduction. Since K is solvable, a non-solvable obstruction above yields one below; conversely, Proposition 3.11 lifts every obstruction below. In the unobstructed case,

$$\mathrm{dl}(H) \leq \mathrm{dl}(\bar{H}) + \mathrm{dl}(H \cap K) \leq \bar{D} + \mathrm{dl}(K)$$

for every proper C-generated H. The opposite inequality $\bar{D} \leq D$ follows from Proposition 3.11. For $r = 2$, $K \cong M_2(\mathbb{F}_p)^+$ is abelian. Finally, all elements of a single conjugacy class have the same determinant, which must generate the unit group if the class generates G. □

Remark 4.5 (A congruence obstruction for a union of classes). In $G = \mathrm{GL}_2(\mathbb{Z}/25)$ put $c = \mathrm{diag}(7, 1)$ and $C = c^G \cup \{6I\}$. Then $H = \langle c^G \rangle = \det^{-1}(\langle 7 \rangle)$ is a proper non-solvable C-generated subgroup of index 5. Indeed, conjugating c by elementary unipotents and multiplying by c^{-1} gives all upper and lower unipotents: the respective coefficients $1 - 7 = -6$ and $1 - 7^{-1} = 8$ are units modulo 25. They generate $\mathrm{SL}_2(\mathbb{Z}/25)$, proving the assertion about H. The determinants 7 and $\det(6I) = 11$ have orders 4 and 5, so C generates G. Its reduction $\bar{C} = \mathrm{diag}(2, 1)^{\mathrm{GL}_2(5)} \cup \{I\}$ is unobstructed. Every non-trivial $\bar{C}$-generated subgroup has full determinant image; every non-solvable subgroup of $\mathrm{GL}_2(5)$ contains $\mathrm{SL}_2(5)$ by Dickson's classification [16, Theorem 3.13], so these two properties force it to be the whole group.

Section 6 gives the rank-one involution classification, followed by the family rates and octahedral estimates in Sections 7 and 8.

5. The synthesis: two solvability conditions

By Theorem 3.2, strict contraction for a configuration is equivalent to its leaves generating G, or escaping every maximal subgroup. This places the matrix and

Group and leaves	Decisive proper subgroups	Exact level / obstruction
S_5, transpositions	S_4 has derived length 3	3
S_n, transpositions, $n \geq 6$	S_{n-1} non-solvable	Obstructed at every depth
A_5, involutions	S_3, D_{10} attain derived length 2	2
$\mathrm{PSL}_2(7)$, involutions	S_4; odd-order 7:3 excluded	3
$\mathrm{PGL}_2(q)$, all involutions, odd $q \geq 5$	$\mathrm{PSL}_2(q)$	Obstructed
$\mathrm{PGL}_2(q)$, exterior involutions	Odd-degree non-solvable subfields; otherwise affine, dihedral or S_4	2, 3 or obstructed (Theorem 6.2)
$\mathrm{PSL}_3(3)$, involutions	Parabolic $3^2{:}\,\mathrm{GL}_2(3)$	5
$\mathrm{SL}_3(3)$, transvections	$3^2{:}\,\mathrm{SL}_2(3)$	4
$\mathrm{GL}_2(\mathbb{Z}_4)$, $\mathrm{GL}_2(\mathbb{Z}_9)$	Ambient groups solvable	Degenerate

TABLE 2. Contraction levels and obstructions for the specified leaf sets.

character obstructions in a common framework: each detects a proper leaf subgroup. For a conjugation-closed generating leaf set in the configurable model, Theorem 3.9 identifies the non-degenerate mixing regime through two conditions:

(i) the ambient group is **non-solvable**, so the commutator tree has a non-trivial accept value (a solvable ambient degenerates, as $\mathrm{GL}_2(\mathbb{Z}_4)$ shows); and
(ii) every **proper C-generated subgroup is solvable**, with the level controlled by the largest derived length among them.

These are exactly the two facts the companion paper [23] noted about S_5 in isolation (S_5 non-solvable; the point-stabiliser S_4 solvable of derived length 3), here a theorem for conjugation-closed leaf sets. Table 2 summarises the verified evidence and the structural obstructions, and Corollaries 5.1, 5.2 and 5.5 record the exact statements.

Corollary 5.1 (Symmetric groups: obstruction at every depth)**.** *For $G = S_n$, $n \geq 5$, with C the class of transpositions, a uniform strict-contraction level exists exactly when $n = 5$, and its exact value is 3 (in the configurable model; Remark 3.7). For every $n \geq 6$ and every depth $m \geq 0$, there is a valid configuration τ with all leaves in the point stabilizer $S_{n-1} < S_n$ whose operator satisfies*

$$\|M_\tau(\rho_{\mathrm{std}})\| = 1,$$

where ρ_{std} is the standard representation of S_n. Consequently no depth gives strict contraction uniformly over all valid configurations for any $n \geq 6$.

Proof. A subgroup generated by transpositions is a Young subgroup $S_{a_1} \times \cdots \times S_{a_r}$, corresponding to the connected components of the transposition graph. For $n = 5$, every proper such subgroup is solvable of derived length at most 3, and the point stabilizer S_4 attains 3; hence Theorem 3.9 gives the exact level.

For $n \geq 6$, let $H = S_{n-1}$ fix the last point. This is a proper non-solvable subgroup generated by transpositions. By Lemma 3.6, the targets of configurations with leaves

in the transposition class of H generate $H^{(m)} \neq 1$ at every depth m. Thus at least one such configuration is valid at each depth. Every leaf fixes

$$v = (1, \ldots, 1, -(n-1)) \neq 0, \qquad v \in \{x \in \mathbb{C}^n : \textstyle\sum_i x_i = 0\},$$

so every leaf projection, and therefore the entire operator, fixes v. As the operator is a product of contractions, its standard-representation norm is exactly 1. This gives the claimed obstruction at every depth. □

Exact shuffles by freely chosen lazy transpositions exist for every S_n [2], with suitable probabilities. Corollary 5.1 identifies when fair independent reads guarantee strict contraction over all valid commutator trees.

Corollary 5.2 (The alternating group A_5)**.** *For $G = A_5$ with C the class of involutions, the construction strictly contracts at exact level* 2 *(configurable class; Remark 3.7).*

Proof. The proper C-generated subgroups are C_2, V_4, S_3 and D_{10}—all solvable, with $\max \mathrm{dl} = \mathrm{dl}(S_3) = \mathrm{dl}(D_{10}) = 2$. (Note A_4 is *not* C-generated: its involutions generate only V_4, illustrating Definition 3.8.) Hence $D = 2$ by Theorem 3.9(2); an exhaustive check confirms no level-1 configuration contracts, while all 29640 ordered valid level-2 configurations (3705 reduced representatives) generate A_5. The numerical bottleneck is approximately 0.346. □

Corollary 5.3 (Spectral characterisation of minimal simple groups)**.** *Let G be a finite non-abelian simple group. The following are equivalent:*
(1) G is minimal simple: every proper subgroup is solvable.
(2) Every non-trivial conjugacy class admits a uniform strict-contraction level in the configurable independent-read model.
(3) Every non-trivial conjugacy class has uniform strict contraction at level 5.
When these conditions hold, the exact level from Theorem 3.9 satisfies $D(G, C) \leq 5$ for every non-trivial conjugacy class C. The universal bound 5 *is attained by the involution class of* $\mathrm{PSL}_3(3)$ *(Proposition 6.4).*

Proof. Every non-trivial conjugacy class C generates G, since $\langle C \rangle$ is non-trivial and normal. If G is minimal simple, all proper C-generated subgroups are solvable and have derived length at most 5 [11, Remark 2.3]. Theorem 3.9 gives the exact-level formula and (1)⇒(3). The implication (3)⇒(2) is immediate.

For (2)⇒(1), suppose that $H < G$ is non-solvable. Choose $x \in H \setminus O_\infty(H)$, where $O_\infty(H)$ is the largest soluble normal subgroup of H, and set $C = x^G$ and $L = \langle C \cap H \rangle$. Conjugation by H preserves $C \cap H$, so $L \trianglelefteq H$. Since $x \in L$, solvability of L would contradict $x \notin O_\infty(H)$. Moreover $L \leq H < G$ and $C \cap L = C \cap H$, so L is a proper non-solvable C-generated subgroup. Theorem 3.9(1) gives an obstruction at every depth, contradicting (2). □

Thus contraction for every non-trivial class singles out precisely the groups in Thompson's classification [20]: $\mathrm{PSL}_2(2^p)$ (p prime), $\mathrm{PSL}_2(3^p)$ (p odd prime), $\mathrm{PSL}_2(p)$ for primes $p > 3$ with $p \equiv \pm 2 \pmod 5$, the Suzuki groups $\mathrm{Sz}(2^p)$ (p odd prime), and $\mathrm{PSL}_3(3)$.

Remark 5.4 ($\mathrm{GL}_3(\mathbb{F}_2)$ has $D = 3$ algebraically). $\mathrm{GL}_3(\mathbb{F}_2) \cong \mathrm{PSL}_2(7)$ is minimal simple, with maximal subgroups two classes of S_4 and one of 7:3, all solvable. For the involution class, the proper C-generated subgroups have maximal derived length $\mathrm{dl}(S_4) = 3$ (the 7:3 is not involution-generated), so $D = 3$ by Theorem 3.9(2). This is an exact algebraic value; the level-3 enumeration in Section 10 only confirms it.

Corollary 5.5 (Projective linear groups)**.** *Let q be odd, $q \geq 5$, and C the set of all involutions of $\mathrm{PGL}_2(\mathbb{F}_q)$ (a union of conjugacy classes, hence conjugation-closed). The involutions of $\mathrm{PSL}_2(\mathbb{F}_q)$ form a single non-trivial conjugacy class of the simple group $\mathrm{PSL}_2(\mathbb{F}_q)$ ($q \geq 5$), hence generate it; adjoining any external involution—for instance the image of $\left(\begin{smallmatrix} 0 & 1 \\ a & 0 \end{smallmatrix}\right)$ with $-a$ a non-square, which squares to aI (hence is an involution in PGL_2) and has determinant $-a$ (a non-square, hence external)—gives $\langle C \rangle = \mathrm{PGL}_2(\mathbb{F}_q)$. Thus the inner involutions generate the proper non-solvable C-generated subgroup $\mathrm{PSL}_2(\mathbb{F}_q)$ (index 2), so by Theorem 3.9(1) this leaf set has no uniform strict-contraction level.* (Restricting C to the PSL_2-external involutions removes PSL_2 from the C-generated subgroups, the exact positive and obstructed cases appear in Theorem 6.2.)

5.1. **Rigid templates: realisable levels and target persistence.** A rigid template fixes one leaf at each position. The per-configuration criterion and the inclusion of level-k targets in $H^{(k)}$ give the upper bound D for every rigid family (Proposition 5.7). We characterise persistence of valid binary continuations through the solvable radical, and show that persistence can be achieved by conjugated doubling (Theorem 5.12 and Corollaries 5.13 and 5.14). Over the transposition class of S_5, the attainable non-vacuous exact levels are 2 and 3 (Theorem 5.15).

Definition 5.6 (Rigid families; non-vacuous exact level)**.** A *rigid family* over a leaf set C is a sequence $F = (F_k)_{k \geq 0}$ of specified sets of level-k templates with leaves in C. Its *formal threshold* $L(F)$ is the least integer $d \geq 0$ such that every valid $\tau \in F_k$, for every $k \geq d$, has $H(\tau) = G$; set $L(F) = \infty$ if no such integer exists. F has *non-vacuous exact level* $D_0 \in \mathbb{Z}_{\geq 0}$ if (i) $L(F) = D_0$; (ii) some $\tau \in F_{D_0 - 1}$ is valid ($\alpha(\tau) \neq e$) with $H(\tau) < G$ (for $D_0 = 0$ this condition is dropped); and (iii) every level $k \geq D_0$ contains a valid member. Condition (iii) rules out vacuous thresholds: a family that is empty above one level may satisfy (i)–(ii) with no content. By Theorem 3.2, $L(F) \leq d$ says exactly that every valid member of level $\geq d$ contracts strictly in every non-trivial irreducible.

Proposition 5.7 (Inherited upper bound)**.** *Let C be conjugation-closed, suppose every proper subgroup generated by a subset of C is solvable, and let D be the maximum derived length among those subgroups (the mixing case of Theorem 3.9). Then $L(F) \leq D$ for every rigid family F over C.*

Proof. Let $\tau \in F_k$ with $k \geq D$ and $\alpha(\tau) \neq e$. By Lemma 3.5, $\alpha(\tau) \in H(\tau)^{(k)}$. If $H(\tau) < G$ then $H(\tau)$ is a proper subgroup generated by a subset of C, hence solvable of derived length at most $D \leq k$, so $H(\tau)^{(k)} = 1$ and $\alpha(\tau) = e$, a contradiction. Hence $H(\tau) = G$. □

Definition 5.8 (Target transition graph)**.** For a finite group G, let $\mathcal{T}(G)$ be the digraph on the non-trivial conjugacy classes of G with an edge $w^G \to (w')^G$

whenever $w' = [u, v]$ for some $u, v \in w^G$. Since $[gwg^{-1}, hwh^{-1}]$ is conjugate to $[w, (g^{-1}h)w(g^{-1}h)^{-1}]$, the graph is computed by sweeping $[w, hwh^{-1}]$ over $h \in G$, one sweep per class.

Lemma 5.9 (Transfer)**.** *If $w^G \to (w')^G$ then for every $u \in w^G$ and $u' \in (w')^G$ there are explicit $g, h \in G$ with $[gug^{-1}, huh^{-1}] = u'$. In particular the relation is well defined on classes, and a self-loop at w^G is equivalent to the existence of a* fixed *pair (g, h) with $[gwg^{-1}, hwh^{-1}] = w$ literally.*

Proof. Write $w' = [g_0 w g_0^{-1}, h_0 w h_0^{-1}]$. If $u = sws^{-1}$, take $g = g_0 s^{-1}$ and $h = h_0 s^{-1}$; then $[gug^{-1}, huh^{-1}] = w'$. Simultaneous conjugation carries w' to any $u' \in (w')^G$. For the loop statement: if $[w, h'wh'^{-1}] = twt^{-1}$ for some t, conjugating by t^{-1} gives $[t^{-1}wt,\ t^{-1}h'wh'^{-1}t] = w$, and both arguments are conjugates of w. □

For a template τ, write $g \cdot \tau$ for the template obtained by conjugating each leaf label s to gsg^{-1}. Its target is $\alpha(g \cdot \tau) = g\alpha(\tau)g^{-1}$.

Theorem 5.10 (Persistence criterion for conjugated doubling)**.** *Let C be conjugation-closed and let τ be a valid template with $H(\tau) = G$ and target $w_0 = \alpha(\tau) \neq e$. Consider continuations by* conjugated doubling*, $\tau_{k+1} = [g_k \cdot \tau_k, h_k \cdot \tau_k]$. The leaves stay in C, and $H(\tau_{k+1}) = G$ whenever $H(\tau_k) = G$. A continuation with all members valid exists if and only if w_0^G reaches a cycle of $\mathcal{T}(G)$; the conjugator schedule may then be taken eventually periodic, with one fixed pair per edge of the cycle.*

Proof. Targets obey $\alpha(\tau_{k+1}) = [\, g_k \alpha(\tau_k) g_k^{-1},\ h_k \alpha(\tau_k) h_k^{-1} \,]$, so the target classes of a valid continuation trace an infinite walk in the finite digraph $\mathcal{T}(G)$, which must reach a cycle. Conversely, fix a canonical representative of each class on a path from w_0^G into a cycle and around it; by Lemma 5.9 each edge is realised by an explicit conjugator pair carrying one canonical representative exactly to the next, so both targets and conjugators are periodic from the moment the cycle is entered. □

We now allow binary compositions with children from different target classes. Fix a conjugation-closed leaf set C; all templates below are over C, so conjugated copies keep their leaves in C. Call a rigid family $(F_k)_{k \geq d}$ a *binary continuation* of F_d if every $\tau \in F_{k+1}$ has the form $[g \cdot \tau_1, h \cdot \tau_2]$ with $\tau_1, \tau_2 \in F_k$ (possibly equal) and $g, h \in G$; call it *persistently valid* if $F_k \neq \emptyset$ *and* every member is valid, for every $k \geq d$. Conjugated doubling is the singleton case. For a set S of non-trivial conjugacy classes let $\Phi(S)$ be the set of non-trivial classes of commutators $[x, y]$ with $x, y \in \bigcup_{C \in S} C$; the diagonal slice ($x, y$ in one class) recovers exactly the edges of $\mathcal{T}(G)$.

Lemma 5.11 (Mixed transfer)**.** *If $[x, y] = z \neq e$ with $x \in A$, $y \in B$, then for every $u \in A$, $v \in B$ and $z' \in z^G$ there are explicit $g, h \in G$ with $[\, gug^{-1},\ hvh^{-1} \,] = z'$: take $g = ca$, $h = cb$ with $aua^{-1} = x$, $bvb^{-1} = y$, $czc^{-1} = z'$.*

Theorem 5.12 (Diagonal reduction)**.** *Let S be a non-empty set of non-trivial conjugacy classes of G and $N(S) = \langle \bigcup_{C \in S} C \rangle$. The following are equivalent:* (1) *$\Phi^n(S) \neq \emptyset$ for all n;* (2) *$N(S)$ is non-solvable;* (3) *some $C \in S$ has non-solvable*

normal closure $\langle C\rangle$, equivalently $C \not\subseteq O_\infty(G)$, the solvable radical; (4) *some $C \in S$ reaches a cycle of $\mathcal{T}(G)$. In particular a class reaches a cycle of $\mathcal{T}(G)$ if and only if its normal closure is non-solvable.*

Proof. (1)⇔(2). $C_0 = \bigcup_{C\in S} C$ is conjugation-closed, so Lemma 3.6 gives $\langle T_k\rangle = N(S)^{(k)}$ for $T_0 = C_0$, $T_{k+1} = \{[a, b] : a, b \in T_k\}$. A commutator with a trivial argument is trivial, and each T_k is conjugation-closed, so induction gives $T_k \setminus \{e\} = \bigcup_{C\in\Phi^k(S)} C$; hence $\Phi^k(S) = \emptyset \iff N(S)^{(k)} = \{e\}$, and (1) holds iff $N(S)$ is non-solvable. *(2)⇔(3).* Each $\langle C\rangle$ is normal (generated by a normal subset) and $N(S) = \prod_{C\in S}\langle C\rangle$; a product of finitely many normal solvable subgroups is solvable, so non-solvability passes to some factor, and conversely each factor lies in $N(S)$. Normality gives $\langle C\rangle$ solvable $\iff C \subseteq O_\infty(G)$. *(3)⇒(4).* If $N_0 = \langle C_0\rangle$ is non-solvable then so is N_0', and Lemma 3.6 (one level, with the class C_0) writes $N_0' = \prod\langle D\rangle$ over the successors $C_0 \to D$ in $\mathcal{T}(G)$; some successor has non-solvable closure, and iterating drives an infinite walk of the finite digraph into a cycle. *(4)⇒(1).* An infinite walk $C_0 \to C_1 \to \cdots$ from $C_0 \in S$ gives $C_n \in \Phi^n(\{C_0\}) \subseteq \Phi^n(S)$ by monotonicity. □

Corollary 5.13 (Binary continuations reduce to doubling)**.** *Let $F_d \neq \emptyset$ be a set of valid level-d templates over C with target-class set S_0. The following are equivalent:* (a) *some binary continuation of F_d is persistently valid;* (b) *$\Phi^n(S_0) \neq \emptyset$ for all n;* (c) *some $\tau \in F_d$ has target class $\not\subseteq O_\infty(G)$;* (d) *some single $\tau \in F_d$ admits a valid conjugated-doubling continuation. Mixed-class children never extend the reach of persistence.*

Proof. (a)⇒(b): the target of $[g \cdot \tau_1, h \cdot \tau_2]$ is $[\, g\alpha(\tau_1)g^{-1}, h\alpha(\tau_2)h^{-1}\,]$, so target classes of valid level-$(d + n)$ members lie in $\Phi^n(S_0)$ by induction. (b)⇒(a): maintain one valid member per class of $\Phi^n(S_0)$, composing by Lemma 5.11 (the same member, doubly conjugated, when the two classes coincide): the children are conjugated copies of valid templates and the root target is a prescribed $z' \neq e$. (b)⇔(c): Theorem 5.12. (c)⇒(d): by Theorem 5.12 the target class reaches a cycle, and the realisation in the proof of Theorem 5.10 uses only the target walk. (d)⇒(a): doubling is a binary continuation. □

Corollary 5.14 (Prefixes admitting an exact-D binary continuation)**.** *In the mixing case of Theorem 3.9 with $D \geq 1$: given $F_0, \ldots, F_{D-1}$ and a non-empty set F_D of valid level-D templates, some binary continuation of F_D yields a rigid family of non-vacuous exact level D if and only if (α) some $\tau^* \in F_{D-1}$ is valid with $H(\tau^*) < G$, and (β) some $\tau \in F_D$ has target class $\not\subseteq O_\infty(G)$. The continuation may then be taken to be conjugated doubling of a single member of F_D; when $O_\infty(G) = \{e\}$, condition (β) holds for every valid member. This characterises binary* extendibility *of a prefix; classifying* prescribed *families remains open (Section 11).*

Proof. $L(F) \leq D$ by Proposition 5.7, whose argument shows every valid member at level $\geq D$ has $H = G$; hence conditions (i)–(ii) of Definition 5.6 reduce to (α) (and $\mathrm{dl}(H(\tau^*)) = D$ automatically, by validity at level $D - 1$). Condition (iii) holds for some binary continuation iff (β): sufficiency is Corollary 5.13; for necessity pass to the subfamily of valid members—subtrees of a valid template are valid, so it is again

a binary continuation, persistently valid by (iii)—and apply Corollary 5.13. The graph data, the transfer identities, the subset-exhaustive equivalence of Theorem 5.12 over nine groups, and a five-level S_5 family with mixed-class children at every production are certified in the accompanying suite. □

Theorem 5.15 (Realisable levels over S_5)**.** *Over the transposition class of S_5, the non-vacuous exact levels realisable by rigid families are exactly* $\{2, 3\}$.

Proof. Exact level 0 is impossible: persistence would require a valid member of F_0, i.e. a single leaf, which generates a group of order $2 < |S_5|$, so $L(F) \geq 1$. Exact level 1 is impossible: persistence would require a valid member of F_1, and a valid pair of transpositions generates $S_3 < S_5$, so $L(F) \geq 2$. And $L(F) \leq 3$ always, by Proposition 5.7 with $D = 3$ (Theorem 3.9, Corollary 5.1).

Level 2 is realised by the family F with $F_0 = \{(12), (23), (34), (45)\}$, $F_1 = \{[(12), (23)], [(34), (45)]\}$ (the leaf configurations and level-1 subtrees of the seed), and $F_k = \{\gamma_k\}$ for $k \geq 2$, where $\gamma_2 = [[(12), (23)], [(34), (45)]]$ and $\gamma_{k+1} = [g \cdot \gamma_k, h \cdot \gamma_k]$ with the fixed pair $g = (4\,5)$, $h = (1\,2)(3\,5)$. The seed generates S_5 with target the 3-cycle $w_3 = (2\,5\,3)$, and $[gw_3g^{-1}, hw_3h^{-1}] = w_3$ (a self-loop of $\mathcal{T}(S_5)$ at the 3-cycle class), so by induction every γ_k, $k \geq 2$, is valid with target w_3 and generates S_5. Hence $L(F) = 2$ (the defining condition holds at $d = 2$ and fails at $d = 1$ through the S_3 members of F_1), the level-1 witness is either member of F_1, and valid members exist at every level ≥ 2: non-vacuous exact level 2.

Level 3 is realised by the family E with E_0, E_1 the leaf configurations and level-1 subtrees of the seed τ_3 below, $E_2 = \{\tau_A, \tau_B\}$ for $\tau_A = [[(12), (23)], [(34), (14)]]$ and $\tau_B = [[(23), (35)], [(45), (24)]]$, and $E_k = \{\tau_k\}$ for $k \geq 3$, where $\tau_3 = [\tau_A, \tau_B]$ and $\tau_{k+1} = [g' \cdot \tau_k, h' \cdot \tau_k]$ with the fixed pair $g' = (3\,4\,5)$, $h' = (2\,3\,4)$. Each of τ_A, τ_B is valid and generates a point stabiliser $S_4 < S_5$; τ_3 generates S_5 with target the 5-cycle $w_5 = (1\,3\,2\,4\,5)$, and $[g'w_5g'^{-1}, h'w_5h'^{-1}] = w_5$, so by induction every τ_k, $k \geq 3$, is valid with target w_5 and generates S_5. Hence $L(E) = 3$ (the condition holds at $d = 3$ and fails at $d = 2$ through τ_A), the level-2 witness is τ_A, and valid members exist at every level ≥ 3: non-vacuous exact level 3.

The accompanying suite certifies these identities and generation facts, and verifies the resulting families through level 6. □

Remark 5.16 (Connection with the companion paper). The persistence criterion supplies valid continuations, and Theorem 3.2 gives qualitative contraction when their leaves generate G. The companion paper [23] provides a quantitative mixing estimate for a specified fixed-template point-function construction. Theorem 5.15 classifies the attainable exact levels across rigid families over the transposition class.

6. Involution classes in rank one

Here the dichotomy gives an explicit classification for an infinite family. Throughout this section $q = p^f \geq 5$ is odd. In $\mathrm{PGL}_2(q)$ the determinant modulo squares defines a character with kernel $\mathrm{PSL}_2(q)$. Call an involution *interior* or *exterior* according as its determinant is a square or a non-square. These are the two involution

classes of $\mathrm{PGL}_2(q)$. The subgroup classification used below is Dickson's, in the form recorded by Giudici [14, Theorems 2.2 and 3.5]; our conclusions concern the commutator-tree levels for these specified leaf sets.

Lemma 6.1 (Classes and subfields)**.** *The interior involutions generate* $\mathrm{PSL}_2(q)$*, and the exterior involutions generate* $\mathrm{PGL}_2(q)$*. Every* A_5 *subgroup of* $\mathrm{PGL}_2(q)$ *is interior. For a proper subfield* $\mathbb{F}_{q_0} \subset \mathbb{F}_q$*, exterior involutions of* $\mathrm{PGL}_2(q_0)$ *remain exterior precisely when* $[\mathbb{F}_q : \mathbb{F}_{q_0}]$ *is odd.*

Proof. The involutions of the simple group $\mathrm{PSL}_2(q)$ form a non-trivial normal subset, so generate it. The subgroup generated by the exterior class is normal in $\mathrm{PGL}_2(q)$. Its intersection with the simple socle is either the socle or trivial; in the latter case its commutator with the socle is trivial, contradicting the trivial centralizer of the socle in $\mathrm{PGL}_2(q)$. Hence it contains the socle and an exterior element, and is the whole group. The determinant character is trivial on the perfect group A_5. Finally, if $m = [\mathbb{F}_q : \mathbb{F}_{q_0}]$, the parity of $(q-1)/(q_0-1) = 1 + q_0 + \cdots + q_0^{m-1}$ is the parity of m. The square-character restriction therefore has the stated parity. □

Theorem 6.2 (Rank-one classification)**.** *For odd* $q \geq 5$ *the following hold, with levels in the configurable model.*

(1) In $G = \mathrm{PGL}_2(q)$*, the set of* all *involutions has a permanent obstruction.*

(2) In $G = \mathrm{PGL}_2(q)$*, the exterior class has a permanent obstruction if and only if there is a proper subfield* $\mathbb{F}_{q_0}$ *with* $q_0 \geq 5$ *and* $[\mathbb{F}_q : \mathbb{F}_{q_0}]$ *odd. Otherwise its exact level is*

$$D = \begin{cases} 3, & q \equiv \pm 3 \pmod 8, \\ 2, & q \equiv \pm 1 \pmod 8. \end{cases}$$

(3) In $G = \mathrm{PSL}_2(q)$*, the involution class has a permanent obstruction if and only if* G *contains a* proper *subgroup isomorphic to* A_5*, or* $\mathbb{F}_q$ *has a proper subfield of size* $q_0 \geq 5$*. Otherwise its exact level is*

$$D = \begin{cases} 3, & q \equiv \pm 1 \pmod 8, \\ 2, & q \equiv \pm 3 \pmod 8. \end{cases}$$

In particular $\mathrm{PSL}_2(5) = A_5$ *has level* 2*. The* A_5 *obstruction in this part occurs exactly when* $q > 5$ *and either* $p = 5$ *or* $q \equiv \pm 1 \pmod 5$.

Proof. For (1), the interior leaves generate the proper non-solvable subgroup $\mathrm{PSL}_2(q)$, so Theorem 3.9 applies.

For (2), a proper exterior-generated subgroup lies in a maximal subgroup other than $\mathrm{PSL}_2(q)$. Giudici's Theorem 3.5 lists these maximals: an affine group $\mathbb{F}_q^+ \rtimes \mathbb{F}_q^\times$, torus normalizers (dihedral groups), S_4 when $q = p \equiv \pm 3 \pmod 8$, and $\mathrm{PGL}_2(q_0)$ when $q = q_0^r$ with r an odd prime. The affine group is metabelian, including over extension fields; its involution-generated subgroups need not be ordinary dihedral groups. Torus normalizers are also metabelian. The only non-solvable possibility outside the socle is therefore a proper subfield group with $q_0 \geq 5$ and odd degree. Such a group is indeed exterior-generated by Lemma 6.1. Any odd proper degree has an odd prime divisor, so the existence criterion agrees with this maximal-subgroup list.

In its absence, the possible subfield maximal with $q_0 = 3$ is $\mathrm{PGL}_2(3) = S_4$, occurring for $q = 3^r$, r an odd prime. Together with the prime-field S_4 case, these give an exterior-generated S_4 exactly in the unobstructed cases $q \equiv \pm 3 \pmod 8$. (Indeed, this congruence forces f odd; without a proper subfield of size at least 5 and odd degree, $f = 1$ when $p \geq 5$, and f is an odd prime when $p = 3$.) Its determinant character is the sign character, so its transpositions are exterior and its derived length 3 is attained. In the remaining cases all relevant maximals are metabelian. The upper bound 2 is attained: for $q \geq 7$, take a split torus generator $r : x \mapsto ax$ and an exterior involution $t : x \mapsto b/x$ ($-b$ non-square). The two exterior involutions t, r^2t generate $\langle r^2, t\rangle$, a proper non-abelian dihedral group whose rotation subgroup has order $(q-1)/2 \geq 3$. The case $q = 5$ has the exterior S_4 and was already covered. Apply Theorem 3.9 for exactness.

For (3), Dickson's list says that any proper non-solvable subgroup lies in an A_5 or a non-solvable subfield subgroup. A proper A_5 is generated by its involutions. Every proper subfield of size $q_0 \geq 5$ supplies a proper involution-generated $\mathrm{PSL}_2(q_0)$, irrespective of the degree; even degree can also supply $\mathrm{PGL}_2(q_0)$. Conversely, the absence of these two obstructions leaves affine and dihedral subgroups of derived length at most 2, and A_4 or S_4. The involutions in A_4 generate only V_4. An S_4 occurs precisely for $q \equiv \pm 1 \pmod 8$ and is generated by involutions, giving level 3 when unobstructed. Otherwise a non-abelian dihedral torus normalizer gives level 2: at least one of $(q-1)/2, (q+1)/2$ is at least 3, and its dihedral normalizer is proper. This argument uses existence, not maximality, of these dihedral groups at small q. At $q = 5$ the A_5 is the ambient group and is not an obstruction, in agreement with Corollary 5.2. The stated A_5 embedding condition is the corresponding part of Dickson's classification. □

Corollary 6.3 (Prime fields)**.** *For every prime $p \geq 5$, the exterior involution class of $\mathrm{PGL}_2(p)$ mixes at exact level 3 for $p \equiv \pm 3 \pmod 8$ and at exact level 2 for $p \equiv \pm 1 \pmod 8$.*

Thus $q = 5, 7, 11$ give exterior levels $3, 2, 3$. At $q = 11$ the involutions of the $\mathrm{PSL}_2(11)$ ambient are instead obstructed by a proper A_5. The extension degree matters: $q = 25$ has exterior level 2, whereas $q = 125$ contains the exterior-generated non-solvable $\mathrm{PGL}_2(5)$ and is obstructed. Section 7 gives mixing rates as q grows.

Proposition 6.4 (An exact level beyond rank one)**.** *For $G = \mathrm{PSL}_3(3)$ and its involution class, the exact level is 5.*

Proof. Here $G = \mathrm{SL}_3(3)$ has trivial centre. Its involutions have eigenvalues $1, -1, -1$, form a single conjugacy class, and generate G by simplicity. The maximal-subgroup list in the ATLAS [21] consists of two parabolic classes $3^2{:}\,\mathrm{GL}_2(3)$, one class of S_4, and 13:3.

Write a point parabolic as $P = V \rtimes L$, where $V \cong \mathbb{F}_3^2$ and $L \cong \mathrm{GL}_2(3)$. In the Levi factor,

$$L' = \mathrm{SL}_2(3), \qquad \mathrm{SL}_2(3)' = Q_8, \qquad Q_8' = \langle z\rangle, \qquad z = -I.$$

The element z acts as negation on V, so $[V, \langle z\rangle] = V$. Consequently the derived series of P is

$$P > V \rtimes \mathrm{SL}_2(3) > V \rtimes Q_8 > V \rtimes \langle z\rangle > V > 1,$$

and $\mathrm{dl}(P) = 5$.

The group L is generated by $s = \mathrm{diag}(-1, 1)$ and the elementary transvections $t = I + E_{12}, I + E_{21}$. Since $sts = t^{-1}$ for either transvection, both s and st are involutions, so L is involution-generated. For every $v \in V$, the product $(vzv^{-1})z$ is the translation $2v$. Thus the involutions of P also generate V, and hence P. The line parabolic has the same properties by inverse transpose.

The remaining maximals are S_4, of derived length 3, and the odd-order group 13:3. Every proper involution-generated subgroup therefore has derived length at most 5, and a parabolic attains 5. Apply Theorem 3.9. □

7. Uniform rates and the order of the mixing depth

The dichotomy supplies generating sets at bounded depth. Diameter estimates then control their contraction uniformly over all valid templates. In particular, the unobstructed exterior classes of $\mathrm{PGL}_2(q)$ have base depth 2 or 3, uniformly in q; the issue here is the strength of contraction as the field grows. All logarithms in this section are natural unless their base is displayed. For a tree τ, the law μ_τ is unconditional.

Theorem 7.1 (Diameter-to-mixing transfer)**.** *Consider a family of pairs (G, C) satisfying case* (2) *of Theorem 3.9, with exact levels $D \le D_0$. Suppose there are constants $K, a > 0$, independent of the pair, such that every symmetric generating set S of G satisfies* $\mathrm{diam}(G, S) \le K(\log|G|)^a$. *Put $c = (8 \cdot 16^{D_0} K^2)^{-1}$. Every valid configurable tree τ of level $m \ge D$ satisfies*

$$d_{\mathrm{TV}}(\mu_\tau, U_G) \le \frac{\sqrt{|G| - 1}}{2} \exp\left(-\frac{c4^m}{(\log|G|)^{2a}}\right). \tag{7.1}$$

For D_T, add the conditioning probability p from Lemma 2.5. The leaves need not be involutions.

Proof. Every sub-block of a valid tree is valid. Its level-D sub-blocks therefore generate G by Theorem 3.9. Each has $N = 4^D$ expanded occurrences, including inverse occurrences. Fix a non-trivial irreducible ρ. For each level-D sub-block τ_0, Proposition 3.17 gives

$$\|M_{\tau_0}(\rho)\| \le 1 - \frac{1}{8 \cdot 4^D K^2 (\log|G|)^{2a}}.$$

Expanding the recursion down to this level produces 4^{m-D} factors, counting adjoints, all with that norm bound. Submultiplicativity and $1 - x \le e^{-x}$ yield

$$\|M_\tau(\rho)\| \le \exp\left(-\frac{4^m}{8 \cdot 16^D K^2 (\log|G|)^{2a}}\right).$$

Apply Diaconis–Shahshahani as in Proposition 3.10, and use $D \le D_0$. □

For $0 < \eta < 1/2$, a sufficient condition for error at most η is $m \geq D$ and

$$4^m \geq c^{-1}(\log|G|)^{2a}\left(\tfrac{1}{2}\log(|G|-1) + \log\frac{1}{2\eta}\right). \tag{7.2}$$

Thus fixed error, or error $|G|^{-b}$ for fixed $b > 0$, requires only $O(\log\log|G|)$ depth and $O((\log|G|)^{2a+1})$ expanded length.

Proposition 7.2 (Support lower bound and optimal order)**.** *For any level-m independent-read tree τ,*

$$d_{\mathrm{TV}}(\mu_\tau, U_G) \geq \max\{0, 1 - 2^{4^m}/|G|\}.$$

Consequently, for any fixed $0 < \eta < 1/2$, in an unbounded family satisfying Theorem 7.1, the least depth at which every valid tree has error at most η is $\Theta(\log\log|G|)$, uniformly over the family.

Proof. There are 4^m independent input bits, hence at most 2^{4^m} possible outputs. The complement of their support has uniform mass at least $1 - 2^{4^m}/|G|$ and output mass zero. Error at most η therefore requires $4^m \geq \log_2((1-\eta)|G|)$. Valid trees exist at every level by Lemma 3.6, since G is non-solvable. This rules out vacuous small levels. The upper bound is (7.2). □

The upper and lower bounds match in their order of depth, up to constant factors. On the obstructed side, the subgroup $H < G$ in Theorem 3.9(1) supplies at every level a valid tree τ supported in H, and hence $d_{\mathrm{TV}}(\mu_\tau, U_G) \geq 1 - |H|/|G| \geq 1/2$.

Corollary 7.3 (Rank one and minimal simple groups)**.** *The uniform rate (7.1) and the optimal order of depth in Proposition 7.2 apply to:*

(1) the unobstructed involution classes of $\mathrm{PSL}_2(q)$ and exterior classes of $\mathrm{PGL}_2(q)$ in Theorem 6.2;

(2) all finite minimal simple groups with any non-trivial conjugacy class, uniformly over both the group and the class.

Proof. Pyber–Szabó's bounded-rank diameter theorem [18, Theorem 2 of the preprint] supplies constants K, a for every symmetric generating set of a simple group of Lie type of bounded rank. For $G = \mathrm{PGL}_2(q)$, take a symmetric generating set S and $t \in S$ outside $L = \mathrm{PSL}_2(q)$. With transversal $\{1, t\}$, Schreier generators of L have length at most 3 in S; their inverses satisfy the same bound. Writing this symmetric set as S_L gives

$$\mathrm{diam}(G, S) \leq 3\,\mathrm{diam}(L, S_L) + 1.$$

Thus the same type of diameter bound holds for G, with adjusted absolute constants. The rank-one levels are at most 3. For (2), Thompson's list in Corollary 5.3 has bounded Lie rank (at most 2, including the underlying type of the Suzuki groups), and Corollary 5.3 gives $D \leq 5$. Apply Theorem 7.1 with $D_0 = 3$ or 5, respectively, and then Proposition 7.2. □

7.1. **Uniform expansion outside exceptional primes.** Over prime fields, strong uniform expansion gives constant gaps for every generating block of bounded depth. The prime-field input goes back to Breuillard–Gamburd [8, Corollary 1.1]. Becker–Breuillard [6, Theorem 1.1 and Corollary 3.2] extend uniform expansion to bounded-rank families with bounded field degree, outside exceptional sets of characteristics.

Corollary 7.4 (Uniform gaps for almost all primes)**.** *There is a set $\mathcal{E}$ of relative density zero among the primes such that the following hold for every prime $p \geq 5$ outside $\mathcal{E}$.*

(1) For each fixed $K \geq 1$ there is $a_K > 0$, independent of p, such that every involutive block τ of level $k \leq K$ whose leaves generate $G = \mathrm{PSL}_2(p)$ or $\mathrm{PGL}_2(p)$ satisfies $\|M_\tau(\rho)\| \leq 1 - a_K$ for every non-trivial irreducible ρ.

(2) Let C be an unobstructed involution class from Theorem 6.2: the exterior class in $\mathrm{PGL}_2(p)$, or the involution class in an unobstructed $\mathrm{PSL}_2(p)$. There is an absolute $c > 0$ such that every valid tree τ of level $m \geq D$ over C satisfies

$$\|M_\tau(\rho)\| \leq e^{-c4^m}, \qquad d_{\mathrm{TV}}(\mu_\tau, U_G) \leq \frac{\sqrt{|G|-1}}{2} e^{-c4^m}. \tag{7.3}$$

The operator bound holds for every non-trivial irreducible ρ.

(3) If $p \equiv \pm 3 \pmod 8$, every pair of exterior subgroups $A, B \cong S_4$ generating $\mathrm{PGL}_2(p)$ satisfies $\|P_A^\rho P_B^\rho\| \leq 1 - b$ for every non-trivial irreducible ρ, with an absolute $b > 0$ independent of the pair and of p.

Proof. Strong uniform expansion supplies $\varepsilon \in (0, 1]$ and $\mathcal{E}$ such that $|SX| \geq (1+\varepsilon)|X|$ whenever $|X| \leq |G|/2$ and S is a symmetric generating set containing e, for both groups G. Starting with [8, Corollary 1.1], pass from $\mathrm{SL}_2(p)$ to its quotient $\mathrm{PSL}_2(p)$ and then to its index-two overgroup $\mathrm{PGL}_2(p)$; uniform expansion is preserved under these operations [6, Lemma 3.9]. Both sources use vertex expansion: the proof of [8, Corollary 1.1] produces a single $s \in S$ with $|sX \setminus X| \geq c|X|$, and [6, (1.1)] states the displayed form directly.

For (1), fix such a block τ and list its $r = 2^k$ leaf positions with their multiplicities. Their lazy walk assigns each distinct non-identity leaf probability at least $1/(2r)$. The expansion inequality gives at least $\varepsilon|X|$ outgoing edges from any X with $|X| \leq |G|/2$, so its conductance is at least $\varepsilon/(2r)$. Cheeger's inequality gives spectral gap $\delta \geq \varepsilon^2/(8r^2)$, including these multiplicities. Each leaf occurs 2^k times in the expanded involutive block, so this is also its expanded-leaf walk. Put $N = 4^k = r^2$. By Proposition 3.19,

$$1 - \|M_\tau(\rho)\|^2 \geq \frac{\varepsilon^2}{4N(N+1)}.$$

Thus one may take $a_K = \varepsilon^2/(8 \cdot 4^K(4^K+1))$.

For (2), $D \leq 3$, and every level-D sub-block of a valid tree τ generates G by Theorem 3.9. Applying (1) at this fixed base level and expanding the recursion gives

$$\|M_\tau(\rho)\| \leq (1-a_3)^{4^{m-D}} \leq e^{-(a_3/64)4^m}.$$

Take $c = a_3/64$ and apply the Diaconis–Shahshahani bound.

For (3), fix a non-trivial irreducible ρ on V, and write $P_A = P_A^\rho$, $P_B = P_B^\rho$. The walk $\nu = (U_A + U_B)/2$ has generating support $A \cup B$ and assigns each support element probability at least $1/48$. The same conductance argument gives spectral gap at least $\eta = \varepsilon^2/(2 \cdot 48^2)$. Its Fourier operator is $(P_A + P_B)/2$, which is positive semidefinite. For a unit vector $v \in V^A$, $1 + \|P_B v\|^2 \le 2(1-\eta)$. Hence $\|P_A P_B\| \le \sqrt{1-2\eta} \le 1-\eta$; if $V^A = 0$ the norm is zero. Take $b = \eta$. □

The all-pairs octahedral programme was formulated in the July 2026 preprint [22, Conjectures 6.9 and 7.8, Remark 7.9], which posed a sharp angular bound and a uniform non-concentration target. Corollary 7.4 gives qualitative gaps for all generating exterior pairs and all ambient-generating level-2 involutive blocks outside $\mathcal{E}$.

7.2. **Few exceptional configurations over arbitrary fields.**

Corollary 7.5 (Exceptional generating blocks at fixed depth)**.** *Fix integers $R, k \ge 1$ and $\delta > 0$. There is $a = a(R,k,\delta) > 0$ such that, for every finite simple group $G(q)$ of Lie type of rank at most R, all but at most $O_{R,k,\delta}(q^\delta)$ simultaneous $G(q)$-conjugacy classes of generating involutive level-k configurations τ satisfy*

$$\|M_\tau(\rho)\| \le 1 - a \qquad \text{for every non-trivial irreducible } \rho.$$

The same conclusion holds for $\mathrm{PGL}_2(q)$, $q \ge 4$. Restricting to valid configurations, or to specified conjugacy classes of involution leaves, preserves this absolute bound on exceptions.

Proof. The dimension-zero theorem of Becker–Breuillard [6, Theorems 1.2 and 4.1], with 2^k entries, gives an expansion constant $\varepsilon = \varepsilon(R,k,\delta) > 0$ outside the stated number of simultaneous conjugacy classes of generating tuples. The finite simple groups have the required bounded-dimensional algebraic embeddings; $\mathrm{PGL}_2(q)$ is covered by Theorem 4.1 in the adjoint group of type A_1, since it contains $\mathrm{PSL}_2(q)$ and therefore satisfies the density hypothesis of that theorem for q large. Finitely many small fields are absorbed into the implied constant. For an involutive tuple put $N = 4^k$. Its lazy leaf walk, with multiplicities, has gap at least $\varepsilon^2/(8N)$ by the conductance argument in Corollary 7.4. By Proposition 3.19 one may take $a = \varepsilon^2/(8N(N+1))$, after reducing ε to at most 1. Conjugating every leaf conjugates the block operator, so the transfer respects exactly the equivalence relation counted by the source theorem. □

Here the depth is fixed and the field degree is unrestricted. The exception count is absolute; a relative-density statement in a restricted family additionally requires a lower bound on its size. The next section obtains the sharp natural-pair constant for every prime in its stated congruence classes.

8. The natural octahedral pair: a sharp uniform angle

We construct a pair of octahedral subgroups whose fixed-space angle is controlled by the Ramanujan bound for a quaternionic Cayley walk. A compression identity connects the two operators. Throughout, q is a prime, $q \equiv \pm 3 \pmod 8$, $q \ne 3$, and

$G_q = \mathrm{PGL}_2(\mathbb{F}_q)$. When a representation ρ is fixed, we write $P_H = P_H^\rho$; we retain the superscript when taking maxima over representations.

8.1. **The pair and its four neighbours.** Let $\mathbb{H} = (-1,-1)_\mathbb{Q}$, with $i^2 = j^2 = k^2 = ijk = -1$, and take projective classes in $\mathbb{H}^\times/\mathbb{Q}^\times$:

$$A_0 = \langle i, 1+i, 1+i+j+k\rangle, \qquad h_0 = i+j+k, \qquad S_0 = \{\pm i \pm j \pm k\}/\{\pm 1\}.$$

The four elements of S_0 have norm 3 and are projective involutions. For any odd prime q, the quaternion algebra over $\mathbb{F}_q$ is split. Indeed, the two sets of squares and of -1 minus squares have more than q elements in total, so $x^2 + y^2 = -1$ has a solution; this gives an explicit splitting into $M_2(\mathbb{F}_q)$ [9, Proposition 2.5.2]. Reducing the indicated projective elements defines A, h, S in G_q. Their norms $1, 2, 4, 3$ are invertible for our primes. Changing the splitting conjugates the entire construction, leaving the norms below unchanged.

Lemma 8.1 (Octahedral reduction)**.** *One has $A \cong S_4$, with exterior transpositions, and*

$$B = hAh^{-1}, \quad A \cap B \cong S_3, \quad h^2 = 1, \quad \langle A, B\rangle = G_q, \quad AhA/A = \{sA : s \in S\}.$$

The last set has four distinct cosets, and S is stable under conjugation by A.

Proof. On the trace-zero subspace with basis (i, j, k), conjugation by the three generators of A_0 and by h_0 gives, respectively,

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad H = \frac{1}{3}\begin{pmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{pmatrix}.$$

The first three generate the 24 determinant-one signed permutation matrices. Their action on the four body diagonals identifies them with S_4; they remain distinct modulo every odd prime. The adjoint action is faithful projectively, also in odd characteristic. Conjugation by $\alpha \in \mathbb{H}(\mathbb{F}_q)^\times$ is $L_\alpha R_{\alpha^{-1}}$ on $\mathbb{H}(\mathbb{F}_q)$, of determinant $N(\alpha)^2 N(\alpha)^{-2} = 1$ for the norm N; it fixes 1 and preserves the norm form, so the adjoint action embeds G_q into $\mathrm{SO}_3(\mathbb{F}_q)$. The norm of $1+i$ is 2, a nonsquare modulo q, so the determinant-square character is non-trivial on A and hence is its sign character. In particular the transpositions of A are exterior.

Here is a uniform intersection calculation. Let K be the stabilizer in A of the line through $(1, 1, 1)$: its matrices are the positive even permutation matrices and the negative odd permutation matrices. Thus $K \cong S_3$, and every element of K commutes with H. Let V be the four diagonal sign matrices of determinant one. Then $A = VK$, and $V \cap K = 1$. The three non-identity elements of V are conjugate under K. For $d = \mathrm{diag}(1, -1, -1)$,

$$HdH = \frac{1}{9}\begin{pmatrix} -7 & -4 & -4 \\ -4 & -1 & 8 \\ -4 & 8 & -1 \end{pmatrix}.$$

Its second row has three nonzero entries modulo every prime $q \geq 5$, so $HdH \notin A$. Since H commutes with K, the same conclusion holds for the other two non-identity cosets of K in A. Therefore $A \cap HAH = K$, in characteristic zero and modulo every

allowed prime. This also proves $h \notin A$ and $B \neq A$. The exterior S_4 is maximal in G_q for these primes by [14, Theorem 3.5]; hence $\langle A, B\rangle = G_q$.

Finally $H^2 = I$, and A acts transitively by conjugation on the four body diagonals S, with stabilizer K. The map $A/K \longrightarrow AhA/A$, $aK \mapsto ahA$, is bijective by the intersection calculation. Since $(aha^{-1})A = ahA$, its images are exactly the four cosets represented by S. □

Proposition 8.2 (The intersection-six stratum)**.** *Let $A_1 \neq A_2$ be exterior octahedral subgroups of G_q. Then $|A_1 \cap A_2| \leq 6$. If $|A_1 \cap A_2| = 6$, the ordered pair (A_1, A_2) is simultaneously conjugate in G_q to (A, hAh^{-1}), and for fixed A_1 there are exactly four such A_2.*

Proof. The exterior octahedral subgroups form one conjugacy class [14, proof of Proposition 3.4], so we may take $A_1 = A$ and $A_2 = gAg^{-1}$ with $g \notin A$, in the orthogonal model of Lemma 8.1.

Normalisers. The three non-identity elements of V have the coordinate axes as fixed lines, so an element normalising V permutes these axes. Orthogonality forces its nonzero entries to be ± 1, and determinant one leaves the 24 elements of A. Thus $N_{G_q}(V) = A$; since V is characteristic in A, also $N_{G_q}(A) = A$.

Orders 8 *and* 12. Put $D = A \cap A_2$. The determinant-square character restricts to the sign character of A and of A_2, so $A \cap \mathrm{PSL}_2(q)$ and $A_2 \cap \mathrm{PSL}_2(q)$ are their A_4 subgroups, whose only subgroups of order 4 are their normal Klein four subgroups. If $|D| = 12$, then D is the A_4 of both groups. If $|D| = 8$, then D contains transpositions, so $D \cap \mathrm{PSL}_2(q)$ has order 4 and is the normal Klein four of both groups. In either case A_2 normalises V, so $A_2 \leq A$, a contradiction. Hence $|D| \leq 6$, and $|D| = 6$ forces $D \cong S_3$.

Order 6. The four S_3 subgroups of A are conjugate in A, so replacing g by $a_1 g a_2^{-1}$ with $a_1, a_2 \in A$, which does not change AgA, we may assume that g normalises K. Every automorphism of S_3 is inner, so $N_{G_q}(K) = K\,C_{G_q}(K)$. The elements of K are the matrices $\mathrm{sgn}(\pi)P_\pi$, $\pi \in S_3$, so a matrix commutes with K exactly when it commutes with every permutation matrix P_π, that is, when it is constant on the two orbits of S_3 on index pairs: the commutant is $\{aI + bJ\}$, with J the all-ones matrix. For $M = aI + bJ$, orthogonality gives $a^2 = 1$ and $b(2a + 3b) = 0$; since $q \neq 3$, M acts by $a + 3b$ on $(1, 1, 1)$ and by a on the complementary plane $x + y + z = 0$, so determinant one gives $a + 3b = 1$. The solutions are I and H. Hence $C_{G_q}(K) = \{1, h\}$, $N_{G_q}(K) = K \sqcup Kh$, and $g \in Kh$ because $g \notin A$. Therefore $AgA = AhA$; writing $g = ahb$ with $a, b \in A$ gives $A_2 = a(hAh^{-1})a^{-1}$, so $(A, A_2) = a(A, hAh^{-1})a^{-1}$.

Count. As $N_{G_q}(A) = A$, the map $gA \mapsto gAg^{-1}$ is injective. The subgroups A_2 with $|A \cap A_2| = 6$ are the images of the cosets in AhA/A, of which there are $[A : K] = 4$ by Lemma 8.1. □

8.2. **Compression and the arithmetic spectral input.** Put $\mu = \frac{1}{4}\sum_{s\in S}\delta_s$ and let e_A denote the uniform average over A in $\mathbb{C}[G_q]$.

Lemma 8.3 (The compression identity)**.** *In the group algebra,*

$$\mu e_A = e_A \mu = e_A h e_A. \tag{8.1}$$

Consequently $\rho(\mu)$ commutes with P_A and its restriction to V_ρ^A is $P_A\rho(h)P_A$, whose norm equals $\|P_AP_B\|$.

Proof. Conjugation invariance of S gives commutation with A. Averaging ahe_A over $a \in A$ visits each of the four cosets in Lemma 8.1 equally often, proving $e_Ahe_A = \frac{1}{4}\sum_{s\in S} se_A$. Apply ρ and Corollary 3.25. □

Proposition 8.4 (Quaternionic Ramanujan input)**.** *For $\chi_q(g)$ the determinant-square character of G_q and $\epsilon_q = (3/q)$, the operator $\rho(\mu)$ has norm at most $\sqrt{3}/2$ on every irreducible $\rho \notin \{\mathbf{1}, \chi_q\}$. On those two characters its values are 1 and ϵ_q.*

Proof. We apply the theorem of Davidoff–Sarnak–Valette [9, Theorem 4.2.2]. Its hypotheses are distinct odd primes p, q with $q > 2\sqrt{p}$, and its distinguished norm-p generators are specified in Section 2.6 of that reference. For $p = 3$ they are exactly the four classes of $\pm i \pm j \pm k$, so every prime here satisfies the hypotheses. Remark 4.4.7 of the same reference derives the Ramanujan bound from Eichler's theorem on weight-two cusp forms.

The vertex set of the walk follows from the construction. The set S is stable under conjugation by A and contains h, so $\langle S\rangle$ is normalised by $\langle A, h\rangle = G_q$ and is a non-trivial normal subgroup of $\mathrm{PGL}_2(q)$. Since $\mathrm{PSL}_2(q)$ is simple with trivial centraliser in $\mathrm{PGL}_2(q)$, $\langle S\rangle$ contains $\mathrm{PSL}_2(q)$. Every $s \in S$ has determinant class 3; hence $\langle S\rangle = \mathrm{PSL}_2(q)$ when $\epsilon_q = 1$ and $\langle S\rangle = \mathrm{PGL}_2(q)$ when $\epsilon_q = -1$. These are the two vertex sets in the cited theorem, and the external input is the spectral bound alone.

If $\epsilon_q = -1$, their Cayley graph on G_q is connected, bipartite and 4-regular, with all eigenvalues other than ± 4 bounded in absolute value by $2\sqrt{3}$. The exceptional normalized eigenspaces are the constants and χ_q. If $\epsilon_q = 1$, the theorem gives a connected nonbipartite Ramanujan graph on $\mathrm{PSL}_2(q)$. On G_q the same walk is the union of two copies, so the exceptional eigenspace at 1 is spanned by the constants and χ_q, and there is no eigenvalue -1. The regular representation contains every irreducible, giving the assertion. In both cases each $s \in S$ has determinant square class 3. □

The compression identity (8.1) identifies the Hecke operator. Left multiplication by the fixed set S on the cosets G_q/A is generally a different operator from the Hecke adjacency. For $q = 5$, those two unnormalized spectra are respectively $\{4, 3, 3, 3, -1\}$ and $\{4, -1, -1, -1, -1\}$. The latter graph is K_5.

Theorem 8.5 (Sharp angle for the natural pair)**.** *For the pair in Lemma 8.1, set*

$$c_q = \max_{\rho\in\widehat{G_q},\,\rho\neq\mathbf{1}} \|P_A^\rho\rho(h)P_A^\rho\|.$$

Then, for every pair A_1, A_2 of exterior octahedral subgroups of G_q with $|A_1\cap A_2| = 6$,

$$\max_{\rho\neq\mathbf{1}}\|P_{A_1}^\rho P_{A_2}^\rho\| = c_q = \max_{\rho\neq\mathbf{1}}\|P_A^\rho P_B^\rho\| \le \frac{\sqrt{3}}{2}, \qquad \lim_{q\to\infty} c_q = \frac{\sqrt{3}}{2}.$$

Thus the constant is best possible uniformly over this family.

Proof. By Proposition 8.2, $(A_1, A_2) = x(A, B)x^{-1}$ for some $x \in G_q$, and $P^{\rho}_{xAx^{-1}} = \rho(x)P^{\rho}_{A}\rho(x)^{-1}$; this gives the first equality in every representation. It remains to treat (A, B). For $\rho \notin \{\mathbf{1}, \chi_q\}$, the compression identity and Proposition 8.4 give the upper bound. On χ_q, $P_A = 0$ because A is exterior, so this character contributes zero.

For the lower bound, define X_q with vertices the left cosets xA, and neighbours xsA for $s \in S$. This is independent of the representative x, since S is invariant under conjugation by A. The four neighbours are distinct and none is xA, by Lemma 8.1. The relation is symmetric because $h^{-1} = h$ and $AhA = (AhA)^{-1}$. The graph is connected because $\langle A, h\rangle = G_q$, and has $|G_q|/24$ vertices. On functions on G_q invariant under right multiplication by A, its normalized adjacency is the restriction of the *right* convolution

$$(Wf)(x) = \frac{1}{4}\sum_{s\in S} f(xs).$$

Under the finite Peter–Weyl decomposition, the multiplicity spaces of this coset representation are V_ρ^A (or their duals, with identical norms), and the blocks are the compressions $P_A\rho(h)P_A$. Hence the absolute nonconstant spectral radius of X_q is $4c_q$. The trivial representation contributes just the constant functions, and χ_q has no A-fixed vectors. Alon–Boppana [9, Theorem 1.3.1], applied to these connected loopless 4-regular graphs with vertex count tending to infinity, gives $\liminf 4c_q \geq 2\sqrt{3}$. Together with the upper bound this proves the limit. □

8.3. **From angles to tree blocks.** The minimum fixed-space angle $\theta_q = \arccos c_q$ satisfies $\theta_q \geq \pi/6$ and $\theta_q \to \pi/6$ by Theorem 8.5. Since $V_\rho^A \cap V_\rho^B = V_\rho^{G_q} = 0$ for every non-trivial irreducible, these are Friedrichs angles, with angle $\pi/2$ when a fixed space is zero. This angular separation controls how two subgroup averages interact. The following estimate transfers it to tree blocks that approximate those averages.

Lemma 8.6 (Quantitative gluing)**.** *Let P, Q be orthogonal projections with $\|PQ\| \leq c < 1$. Suppose $X = P + R$, $Y = Q + S$, where $PR = RP = QS = SQ = 0$ and $\|R\|, \|S\| \leq \lambda < 1$. Then*

$$\|XYX^{\dagger}Y^{\dagger}\|^2 \leq 1 - \frac{(1-\lambda)(1-c)}{8(1+\lambda)}. \tag{8.2}$$

Proof. Put $g = 1 - \lambda^2$. Each factor Z with associated projection F satisfies

$$\|w\|^2 - \|Zw\|^2 \geq g\|(I-F)w\|^2, \qquad \|w - Zw\| \leq (1+\lambda)\|(I-F)w\|.$$

Apply the four factors in their actual order to a unit v, and denote their norm-squared losses by $e_j \geq 0$. Their sum is $E = 1 - \|XYX^{\dagger}Y^{\dagger}v\|^2$. At each step, telescoping back to v and Cauchy–Schwarz give

$$\|(I-F)v\| \leq \frac{1+\lambda}{\sqrt{g}}\sum_{i\leq j}\sqrt{e_i} \leq \frac{2(1+\lambda)}{\sqrt{g}}\sqrt{E}.$$

Both P and Q occur. Also $\|P+Q\| \le 1+c$: the map $v \mapsto (Pv, Qv)$ has squared norm bounded by that of the block matrix with identity diagonal and off-diagonal blocks PQ, QP. Consequently

$$1 - c \le \|(I-P)v\|^2 + \|(I-Q)v\|^2 \le \frac{8(1+\lambda)^2}{g} E.$$

Taking the supremum over v proves (8.2). □

Corollary 8.7 (Uniform contraction for natural-pair blocks)**.** *Suppose a valid level-3 tree τ has level-2 leaf subgroups $A, B \cong S_4$, exterior with $|A \cap B| = 6$, and all leaves are exterior involutions. For every non-trivial ρ,*

$$\|M_\tau(\rho)\| \le \kappa, \qquad \kappa = \left(1 - \frac{183}{1736}\left(1 - \frac{\sqrt{3}}{2}\right)\right)^{1/2} < 1.$$

For a valid level-m tree, $m \ge 3$, all of whose level-3 sub-blocks have this property, replace λ by κ and D by 3 in Proposition 3.10.

Proof. On the fixed space of its leaf subgroup, each level-2 operator is the identity. On the orthogonal complement its norm is at most $\lambda = 17/200$: this is a finite rational certificate on the four non-trivial irreducibles of S_4, over its 432 valid ordered transposition configurations generating S_4. The exact certificate described in Section 10 constructs rational matrices with invariant Gram matrices G_π and verifies all principal minors of $(17/200)^2 G_\pi - D^t G_\pi D$ are nonnegative. Restriction of any ambient unitary representation to S_4 decomposes into these irreducibles and the trivial one; hence the certificate applies in every ambient G_q. Both invariant subspaces are preserved, so the hypotheses of Lemma 8.6 hold. Insert $c = \sqrt{3}/2$ from Theorem 8.5. The last assertion follows by expanding into the level-3 blocks, as in Proposition 3.10. □

The sharp angle estimate and the finite S_4 certificate thus give the explicit contraction bound κ for these octahedral blocks. Applying Proposition 3.17 inside S_4 also gives a purely analytic, weaker constant in the same corollary.

9. Relation to word maps, conjugacy walks, and expansion

The following connections identify the standard inputs and the questions addressed here.

Word maps. Word maps connect random inputs in finite groups to questions about images and distributions [15]. For any fixed iterated commutator in distinct variables and any fixed arrangement of brackets, Garion–Shalev [13] prove asymptotic equidistribution on finite non-abelian simple groups as their orders tend to infinity. Their variables are uniform group elements, and repeated occurrences of a variable use the same value. Here each expanded occurrence contributes one independent fair bit: the operator uses $XY\tilde{X}^{-1}\tilde{Y}^{-1}$ with independent copies (Lemma 2.2 and Remark 2.3). At fixed depth its output support has size at most 2^{4^m}, forcing depth $\Omega(\log\log|G|)$ for fixed-error mixing. Theorem E attains this order in the specified families.

Conjugacy-class random walks. When the leaf set is a single conjugacy class C, uniform averaging over C has eigenvalues $\chi(C)/\chi(1)$, with a classical mixing theory [10, 17]. In the commutator tree, the leaf-subgroup criterion detects trapping through fixed spaces in $\mathrm{Ind}_H^G \mathbf{1}$. The non-identity target then forces escape from proper solvable leaf subgroups at the depth given by Theorem 3.9.

Expansion and solvability. Bourgain–Gamburd [7] and Salehi Golsefidy–Varjú [19] establish expansion under hypotheses on generators, girth or the connected component of the Zariski closure, depending on the theorem. In the tree model, a common fixed vector for a proper leaf subgroup is the exact obstruction to strict contraction. For bounded-depth involutive blocks, Corollary 3.20 relates a uniform gap to expansion of their particular leaf walks. Its upper comparison uses Gao's projection union bound as presented in [3]. The family rates of Theorem E use the bounded-rank diameter theorem [18] and allow the fixed-depth gap to decrease with group size. Strong uniform expansion gives the prime-field refinement in Corollary 7.4.

Mixability of free subproducts. Closest is the *mixability* programme of Amir, Blachar, Ghosh and Vishne [1]: when is a free random subproduct $g_1^{\varepsilon_1} \cdots g_k^{\varepsilon_k}$ (arbitrary g_i, independent Bernoulli ε_i) *exactly* uniform? They characterise it—conjecturally—by $2'$-simplicity (no non-trivial odd-order quotient), through a representation criterion $\prod_i \big((1-p_i)I + p_i\rho(g_i)\big) = 0$ whose singular-step lemma is a linear cousin of our Lemma 3.1. Their steps and Bernoulli parameters may be chosen freely. Here the expanded steps are constrained by a balanced tree whose folded target must be non-trivial. This validity condition brings the derived series into the threshold. In particular, their mixability of every S_n concerns different choices of steps from our uniform statement over valid transposition trees.

Structural conclusion. "Mixing $\iff$ the steps generate G" is classical for random walks. For the independent-read commutator tree, the resulting structural statement is: the *exact uniform obstruction* is a proper non-solvable C-generated subgroup, and the *exact level* is the largest derived length among proper C-generated subgroups (Theorem 3.9).

10. Computational verification

The artefact contains exact group computations, rational norm certificates, and numerical checks. At level 2, the tables enumerate reduced representatives: interchanging leaves within either gate and interchanging the gates preserves validity and the generated subgroup. Each valid orbit has eight ordered tuples. These reductions certify validity and subgroup generation. Certified numerical upper bounds on norms use exact checks covering all orientations; NumPy singular values serve as cross-checks. The S_5 certificate is supplied by the companion paper [23]. The rank-one classification and the level-5 example are proved algebraically. The script `psl33_maximals_check.py` independently checks the orders, involution generation and derived lengths used in Proposition 6.4.

The natural octahedral pair. `octahedral_pair_check.py` checks exact subgroup orders, intersections, generation, coset adjacency and the compression identity

for $q = 5, 11, 13, 19$; it distinguishes the two coset operators exactly at $q = 5$. Its larger spectral checks are numerical. The supplementary Hamilton reduction check uses exact integers. The script `s4_internal_gap_certify.py` supplies the rational certificate used in Corollary 8.7, checking all principal minors for the 432 valid ordered S_4-generating transposition configurations and all four non-trivial irreducibles. `octahedral_cap6_check.py` verifies, for $q = 5, 11, 13, 19, 29, 37$, the intersection census of Proposition 8.2, the uniqueness of the double coset of intersection order six with its four neighbours, the normaliser and centraliser computations, and the orthogonal dictionary. It certifies the natural values c_q exactly: the double-coset quotient of the Hecke graph carries every distinct eigenvalue, and its integer characteristic polynomial is factored over $\mathbb{Z}$ with rational root isolation (SymPy). For $q = 11, 13, 19$ this gives $c_q = (1 + \sqrt{3})/4$, $3/4$, $(1 + \sqrt{5})/4$; the certificates for $q = 29, 37$ are recorded in the artefact. Arbitrary-order stability and the support bound also have small-group numerical regression tests.

All-pairs numerical evidence. For an exterior S_4 subgroup $A < \mathrm{PGL}_2(q)$ and each non-trivial double coset AgA, the non-trivial singular norm of the coset operator on $\ell^2(G_q/A)$ equals that of its restriction to A-invariant functions, an orbit-weighted matrix indexed by $A\backslash G_q/A$. `octahedral_cap6_check.py` computes these norms by one floating-point procedure for every prime in Table 3, deflating a single copy of the trivial singular value, and cross-checks them against dense singular value decompositions on $\ell^2(G_q/A)$ for $q \le 19$. The values are numerical. In each case the maximum is attained at $|A \cap gAg^{-1}| = 6$ and agrees, to the precision shown, with the separately certified natural value c_q. The historical rank-one experiments [22], bundled with provenance as `external/p2a/s4_allpairs_doublecoset_audit.py`, estimated the same quantities by power iteration.

q	11	13	19	29	37
All-pairs maximum	0.683013	0.750000	0.809017	0.854361	0.845651

TABLE 3. Numerical all-pairs maxima, compared with $\sqrt{3}/2 \approx 0.866025$.

Transvection levels. `transvection_levels.py` independently cross-checks the $\mathrm{SL}_3(3)$ case of Proposition 4.2 by enumerating transvection-generated subgroups up to conjugacy in exact arithmetic. It finds 104 transvections and eight non-trivial subgroup classes including G; the largest proper derived length is 4, attained at order 216.

The generation criterion (reduced level-2 universe). For each reduced valid configuration we compare the numerical regular-representation second singular value with its exact leaf-subgroup closure, as predicted by Theorem 3.2. The counts in Table 4 are reduced; the respective ordered totals are 105168 and 219360.

Mixing levels. Table 5 lists the level at which contraction sets in, with its certification status. The S_5 five-point row reproduces the companion paper [23] to the digit for the reduced enumeration ($\lambda \approx 0.0353047532$ over 73810 reduced level-3 configurations). The corrected full oriented enumeration has maximum $\lambda \approx 0.03540023208379892$,

Group	Reduced	Intransitive	Proper	Generate G	Mismatches
$\mathrm{GL}_3(\mathbb{F}_2)$	13146	2772	2772	10374	0
S_5 on 6 pts	27420	0	4695	22725	0

TABLE 4. Reduced level-2 all-involution checks. "Intransitive" counts augmentation norm one; "Proper" counts regular second singular value one, agreeing with the exact leaf closure in every case. The augmentation test misses 4695 proper transitive closures for S_5 on six points (3705 of order 60, 990 of order 24).

Action and leaves	Stabiliser / dl	Level and quantitative evidence
S_5, 5 points, transpositions	S_4 / 3	3; exact bound 0.0355 [23]
S_5, 6 points, all involutions	F_{20} / 2	2 in augmentation only; full contraction obstructed
$\mathrm{PGL}_2(7)$, 8 points, exterior	Borel / 2	2 analytically; norms sampled
$\mathrm{GL}_3(\mathbb{F}_2)$, 7 points, involutions	S_4 / 3	3 analytically; level-3 norms capped
S_6, 6 points, transpositions	S_5 / non-solvable	No uniform level; obstructed configurations persist

TABLE 5. Contraction levels and quantitative evidence.

so the bound $71/2000 = 0.0355$ is certified in exact integer arithmetic by `20_level3_exact_certificate.py` from [23], supplied unchanged in the artefact with its provenance. It covers 9447840 oriented pairs through 39591 symmetry orbits; the sign representation vanishes separately.

A degenerate example. $\mathrm{GL}_2(\mathbb{Z}_4)$ produces zero valid level-3 configurations, confirming the degeneracy of Section 4.

11. Open problems

(1) **Uniform expansion and sharp angles.** For prime $q \equiv \pm 3 \pmod 8$, $q \neq 3$, is

$$\sup_{\substack{A,B\cong S_4 \text{ exterior} \\ \langle A,B\rangle = \mathrm{PGL}_2(q)}} \sup_{\rho\neq \mathbf{1}} \|P_A^\rho P_B^\rho\| \leq \sqrt{3}/2 + o(1) \qquad (q \to \infty)?$$

This question was formulated in the preliminary rank-one preprint [22, Conjecture 6.9]. Proposition 8.2 and Theorem 8.5 prove it for every pair with $|A \cap B| = 6$, so it remains open for the intersection orders $1, 2, 3, 4$; Table 3 records finite numerical evidence. A separate question is whether the qualitative gaps of Corollary 7.4 for all exterior generating pairs and all ambient-generating level-2 involutive blocks extend to every admissible prime. Finally, do the involution classes of the minimal simple families $\mathrm{PSL}_2(2^\ell)$, $\mathrm{PSL}_2(3^\ell)$ and $\mathrm{Sz}(2^\ell)$ admit a constant gap for every valid level-D block, uniformly as the prime exponent ℓ grows (odd in the latter two families)? Corollary 7.5 allows exceptional configurations; removing them in these families is a concrete consequence

sought from the bounded-rank uniform expansion conjecture discussed in [6, Introduction].

(2) **Fixed templates.** Characterise which *prescribed* rigid families attain the configurable bound D: Proposition 5.7 gives $L(F) \leq D$ for all of them, Theorem 5.15 shows that over S_5 both D and $D-1$ occur, and Corollary 5.14 characterises prefixes admitting an exact-D continuation. For a prescribed family, determine the optimal contraction rate and whether its base-level gap stays positive as the ambient group varies. Submultiplicativity propagates a given base gap; the sharp rate remains to be determined.

(3) **Irreducible matrix classes and congruence levels.** For $n \geq 3$, classify the generating conjugacy classes of $\mathrm{GL}_n(\mathbb{F}_q)$ with irreducible characteristic polynomial for which every proper class-generated subgroup is solvable. These elements preserve no proper nonzero subspace, so this question concerns the non-parabolic maximal subgroups. Proposition 4.2 settles the contrasting transvection family in $\mathrm{SL}_n(q)$. For a generating conjugacy class of $\mathrm{GL}_2(\mathbb{Z}/p^2)$, $p \geq 5$, whose reduction is unobstructed, determine exactly when the level in Proposition 4.4 is $\bar{D}$ or $\bar{D}+1$. More generally, characterise the proper class-generated supplements to congruence kernels when unions of classes are allowed, starting from Remark 4.5.

AI Declaration

OpenAI ChatGPT/Codex, Anthropic Claude and Google Gemini were used to search for counterexamples and errors in proofs, assist with developing and verifying code, and edit the language. The author is responsible for the mathematical statements, proofs, references, code and exposition.

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Conflict of interest. The author declares no conflict of interest.
Data access. The verification artefact and its sources are publicly deposited at `https://github.com/ernest-zermeno/commutator-tree-mixing-finite-groups`. Its README and manifest identify exact certificates, numerical checks, sampled runs and the certificate from [23].

For Section 5.1, `bridge_realizability_certificate.py` checks the S_5 rigid families, `target_transition_graph.py` checks target transitions, and `binary_target_grammar_certificate.py` checks the binary persistence criterion in the groups listed in its documentation.

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